

MATERIALS SCIENCE

3D-Printed biodegradable and multifunctional cranial device for bone repair and healing assessment

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Critical-sized bone defects pose substantial clinical challenges, requiring both regenerative and diagnostic strategies for effective treatment. Here, we report a biodegradable, multifunctional cranial device based on three-dimensional printing. The device integrates a self-electrified scaffold based on a transient zinc-molybdenum battery and an impedance sensor for simultaneous electrical stimulation and continuous assessment of bone healing. The beneficial effects of electrical cues are attributed to elevated calcium ion influx, reactive oxygen species signaling, and up-regulated osteogenesis-related genes. In a rodent cranial defect model, the device promotes osteogenesis while enabling dynamic tracking of early tissue regeneration through impedance measurements. The increase in impedance is associated with enhanced collagen deposition and matrix mineralization. By uniting therapeutic stimulation with in situ monitoring, this work offers a platform for intelligent bone repair with broad potential in regenerative medicine.

INTRODUCTION

Bone defects resulting from trauma, surgical resection, or infection necessitate extensive surgical intervention, imposing a global burden of physical deformity, diminished quality of life, and high treatment costs (1). While minor bone injuries can heal through intrinsic regenerative capacity, large defects often exceed this capability, precluding functional recovery without surgical intervention (2). Autologous bone grafting, the clinical gold standard, is constrained by donor scarcity, size mismatch, and the risk of secondary morbidity (3). Synthetic scaffolds therefore offer a promising alternative by mimicking the native extracellular matrix (ECM) to provide structural support for cellular adhesion, proliferation, and osteogenic differentiation (4). However, most pure scaffolds lack inherent bioactivity, limiting their

regenerative efficacy. Electrical cues represent an additional bioactive functionality that has been clinically used to promote bone repair, demonstrating efficacy in managing delayed union and nonunion fractures (5). It acts by directing osteoprogenitor cell migration, promoting osteogenic differentiation, and facilitating the deposition of extracellular matrices (such as mineralization and collagen formation) (6). The beneficial effects of electrical stimulation have been attributed to the up-regulation of intracellular calcium (Ca^{2+}) levels and the expression of regeneration-associated proteins, such as bone morphogenetic proteins (Bmp2), which together promote osteogenic differentiation and matrix mineralization (7). In addition, electrical cues can modify the local microenvironment (such as oxygen tension and pH) and trigger intracellular responses by inducing hydroxide ions (OH^-) and/or reactive oxygen species (ROS) generation at the wound sites, which can activate mitogen-activated protein kinase cascades and up-regulates transforming growth factor- β (TGF- β) and Bmp2 expression to promote osteogenesis (8, 9).

Conventional electrical stimulation strategies depend on wired connections to external power sources, delivering either direct or alternating currents, but their tethered nature constrains application and introduces infection risks associated with percutaneous leads. Wireless alternatives—including piezoelectric electronics (10, 11), triboelectric nanogenerators (7), magnetoelectric systems (12), and optoelectronic platforms (13)—have been developed to overcome these challenges. However, these approaches remain contingent on externally applied stimuli, such as light, magnetic fields, or mechanical actuation, which complicates implementation in vivo. These limitations underscore a critical unmet need for self-electrified devices that can autonomously generate therapeutic electrical cues, an approach that remains largely unexplored for bone regeneration. Moreover, real-time assessment of bone healing is critical for early diagnosis and timely clinical intervention; yet, this capability remains absent in current therapeutic systems. At early stages of healing, mineralization is incomplete, severely limiting the sensitivity of conventional imaging techniques such as radiography and computed tomography (CT), which depend heavily on detectable mineral deposition (14).

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An ideal biointegrated device would autonomously deliver osteogenic electrical cues while simultaneously monitoring the early regeneration process, offering a closed-loop therapeutic strategy. Furthermore, endowing such a device with biodegradability would avoid secondary surgical removal, minimize long-term foreign-body retention, and reduce infection risks. Despite this desirable clinical need, a biodegradable, multifunctional bioelectronic system that combines autonomous electrical stimulation with real-time healing monitoring has not yet been realized.

In this study, we present a biodegradable, electronic cranial device based on three-dimensional (3D) printing that uniquely integrates a self-electrified scaffold leveraging a zinc-molybdenum (Zn-Mo) battery with bioimpedance-monitoring electrodes, enabling simultaneous delivery of electrical cues for bone repair and in situ real-time assessment of healing. The scaffold, composed of hydroxyapatite (HA) mineralized collagen/polycaprolactone (MC/PCL) and a 3D-printed Zn-Mo battery. The autonomous electrical signal triggers intracellular calcium flux and modulates ROS generation, promoting osteogenic differentiation of stem cells in vitro. In our system, the Zn-Mo battery generates an electrical field that drives a small electric current through the surrounding tissues, and the term “electrical stimulation” refers to the battery-induced electric field and current. In a rat cranial defect model, the self-electrified scaffold accelerates bone repair by activating TGF- β and ECM-receptor signaling pathways, which up-regulate osteogenic, angiogenic, and ECM-related genes. In parallel, integrated Mo/poly(L-lactide-co- ϵ -caprolactone) (Mo/PLCL) electrodes enable localized monitoring of tissue regeneration during the critical early stages of healing, effectively distinguishing between successful and impaired bone repair. The entire device undergoes complete biodegradation under physiological conditions, thereby eliminating the need for secondary surgical removal. Together, these features establish a multifunctional, self-electrified platform that represents a promising and intelligent strategy for advanced cranial reconstruction.

RESULTS

Materials strategies and device fabrication

Figure 1A shows a schematic illustration of the biodegradable and multifunctional cranial device, comprising two components: (i) a self-electrified scaffold consisting of a 3D-printed Zn-Mo galvanic battery sandwiched between two MC/PCL layers, to provide electrical cues for promoting bone repair, and (ii) 3D-printed Mo/PLCL electrodes for monitoring bioimpedance. We evaluate the Young's modulus and compressive strength of the MC/PCL scaffold, and the result is given in fig. S1. The scaffold has a Young's modulus of 1.10 ± 0.27 MPa and a compression strength of 1.00 ± 0.52 MPa, which is close to the mechanical properties of the cancellous bone (15). The scaffold is expected to provide adequate structural support for cranial defects, where the load-bearing requirement is moderate. The direct ink writing (DIW) 3D-printing technique is used to fabricate the cranial device because of its advantages in customizing geometry and integrating multifunctional components. Zn-Mo galvanic batteries and Mo/PLCL electrodes are printed using cellulose nanofibril (CNF)-based ink and PLCL-based ink, respectively (Fig. 1A). For the self-electrified scaffold, biodegradable MC/PCL (~ 200 μ m) serves as the top and bottom substrate (fig. S2A) to mimic the porous architecture and composition of natural bone. 3D-printed Zn and Mo electrodes are used as the anode and cathode of the galvanic cell because of their appropriate electrochemical potentials,

proven biodegradability, and suitable operational lifespan in physiological environments (16). Zn ions have also been reported to be osteogenic and are therefore beneficial for bone repair at appropriate concentrations (17). Moreover, the biodegradable Mo/PLCL ink is 3D printed onto a PLCL thin film (~ 200 μ m) to create flexible electrodes for impedance monitoring. The self-electrified scaffold and Mo/PLCL electrodes are subsequently integrated into a multifunctional electronic device via solvent bonding and deployed at cranial defects in rodents, to promote bone regeneration and enable healing assessment (Fig. 1A and fig. S2B).

Currently available 3D printing inks for electronics mostly rely on nondegradable constituents, such as silver, gold, carbon nanotubes, polyaniline, polypyrrole, and poly(3,4-ethylenedioxythiophene) polystyrene sulfonate (18–20). Therefore, the development of fully biodegradable inks is essential, particularly to meet the compositional and rheological requirements of the DIW printing process, which demands smooth extrusion through fine nozzles, strong interlayer adhesion, and structural integrity postdeposition (21, 22). Here, we develop 3D-printable conductive inks with favorable rheological properties and extrusion performance by incorporating biodegradable CNFs with Zn and Mo (Zn/CNF and Mo/CNF) particles, and PLCL with Mo particles (Mo/PLCL). In addition, inks composed of CNFs, alginate (ALG), and carboxymethyl cellulose (CMC) are explored for use as gel electrolytes (ALG/CMC/CNF). CNFs with a high aspect ratio form a robust, self-supporting network through hydrogen bonding and physical entanglement, conferring structural integrity without deformation after printing process (Fig. 1A). Under shear, disruption of these hydrogen bonds induces shear-thinning behavior, a key characteristic for extrusion-based printing. The addition of metal particles at an optimal solid content further enhances printability by reinforcing the printed architecture and ensures adequate conductivity. Moreover, as a rubber-like biodegradable copolymer, PLCL provides excellent elasticity and printability, and serves both as a binder for conductive fillers and as a matrix material, imparting desirable flexibility to the printed electrodes.

All inks exhibit strong shear-thinning behavior typical of non-Newtonian fluids, as evidenced by the decrease in apparent viscosity with increasing shear rate (Fig. 1B). While the CNF-based inks share similar shear-thinning characteristics, the Mo/PLCL ink displays a lower apparent viscosity, facilitating easier extrusion. Oscillatory rheology measurements reveal the storage modulus (G') and loss modulus (G''), with yield points observed at 39.8, 10.4, 15.9, and 6.3% strain for Mo/CNF, Zn/CNF, ALG/CMC/CNF, and Mo/PLCL inks, respectively (Fig. 1C). As strain increases, both G' and G'' decline, indicating a viscoelastic transition from elastic-dominant to viscous-dominant behavior near the yield points. The shear-thinning nature of the matrix facilitates extrusion-based printing, while its viscoelastic transition preserves structural integrity by preventing gravity-induced deformation of the printed structure. The particle size distributions of Zn and Mo are evaluated, and the results are given in fig. S3. The results indicate a reasonable distribution range. Although excessively large particles can cause print clogging, a common guideline for stable inkjet printing is that the particle size should be at least 50 times smaller than the nozzle diameter, which helps prevent issues such as particle accumulation at the nozzle edge, deviations in droplet trajectories, and clogging due to agglomeration (23, 24). Given that our printing nozzle has a diameter of 410 μ m, the observed particle size distributions fall well within the acceptable range for stable printing.

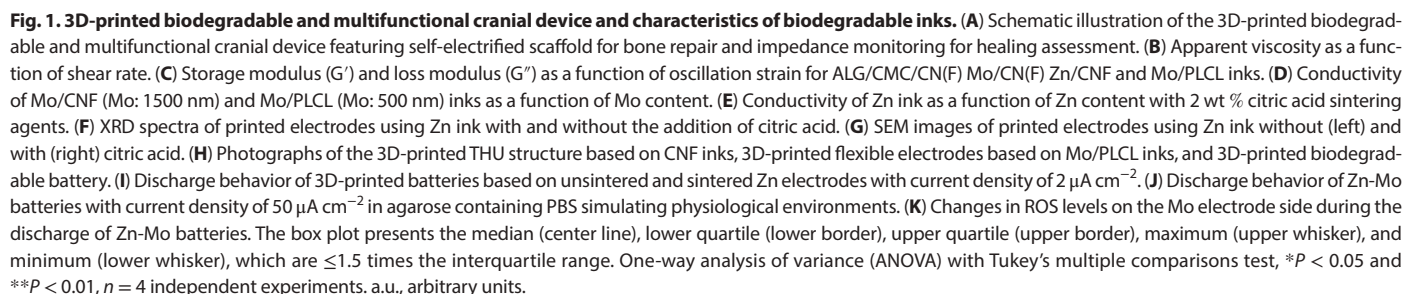


Figure 1D summarizes the conductivity of Mo/CNF ink (Mo particle size: 1500 nm) and Mo/PLCL ink (Mo particle size: 500 nm) as a function of Mo particle loading. For Mo/CNF inks, conductivity increases slightly with rising Mo content, peaking at approximately 25 kS m^{-1} at ~50 to 55 wt %. Beyond this point, additional Mo results in a decrease in conductivity. A similar trend is observed in Mo/PLCL inks, which reach a maximum conductivity of 7.3 S m^{-1} at 50 wt % Mo loading, followed by a decline at higher concentrations. This is attributed to the reduced polymer content, which results in inhomogeneous dispersion of Mo particles within the matrix, thereby affecting conductivity, a phenomenon consistent with previous reports (25, 26). Moreover, when the solid particle content exceeds 50 wt %, the high concentration of metal particles disrupts filament continuity after extrusion (fig. S4). Therefore, a solids content of 50 wt % is identified as optimal for ensuring printability, structural fidelity, and desirable conductivity. The relationship between ink conductivity and the volume fraction of Mo particles is also presented in fig. S5A, with 10% v/v identified as the optimal fraction. Although the Mo/PLCL composite exhibits moderate electrical conductivity, this value lies within the range reported for nondegradable 3D-printed electrodes used in bioelectronic sensing applications (0.1 to 10^5 S m^{-1}) (18, 27). Moreover, the impedance of the Mo/PLCL electrodes at 10 kHz is approximately 16 k Ω , which falls within the reported range for neural electrodes (0.519 to $10^3 \text{ k}\Omega$ at 1 kHz) (28). As will be shown later, during bone regeneration, mineralization of collagen markedly increases tissue impedance by several orders of magnitude, which is much greater than the impedance of the electrodes, indicating that the Mo/PLCL electrode can enable reliable monitoring of bone tissue impedance through the healing process in a physiological environment.

In contrast, the conductivity of Zn inks is orders of magnitude lower than that of Mo inks because of the spontaneous surface oxidation of Zn particles, which forms a thin zinc oxide (ZnO) layer that degrades particle connectivity. Treatment with mild acids can uniformly remove the oxide layer, improve interparticle connectivity, and thereby enhance electrical conductivity, an effect commonly referred to as sintering (16, 29). We introduce four weak acid sintering agents into the ink formulation, including propane sulfonate lactone, citric acid, gluconolactone, and aspartic acid (fig. S6). Although propane sulfonate lactone results in higher conductivity, its potential biotoxicity and rapid reaction kinetics that causes nozzle clogging, limits its practical use. Citric acid is ultimately selected as the optimal sintering agent because of its effective sintering performance and favorable biocompatibility. Similar to Mo/CNF, Zn/CNF with solid contents below 50 wt % is preferred to ensure printability, while the highest conductivity is achieved with the addition of 2-wt % citric acid at 50-wt % solid content (fig. S7). Higher concentrations of citric acid reduce conductivity, likely because of the formation of slightly soluble zinc citrate deposits on Zn particle surfaces, which impedes the formation of conductive pathways. The conductivity of Zn/CNF inks formulated with Zn particles of three different sizes (400, 700, and 2600 nm) is also evaluated with the addition of 2-wt % citric acid (Fig. 1E and fig. S5B). Zn particles with a size of 700 nm result in slightly higher conductivity than those with a 400-nm size and notably better conductivity than the 2600-nm particles, for its better dispersibility and capacity to form a continuous conductive network. A conductivity of $\sim 0.2 \text{ S m}^{-1}$ is achieved at 50 wt % and 14% v/v using 700-nm Zn particles. Therefore, considering both printability and conductivity, Zn/CNF inks containing 50-wt % Zn

with a particle size of 700 nm and 2-wt % citric acid are selected for further studies. Although moderate, the conductivity of Zn/CNF falls within the reported range (0.01 to 260 S m^{-1}) for nondegradable inks used in 3D-printed batteries or supercapacitors that exhibit its desirable performance (30–32).

X-ray diffraction (XRD) analysis of Zn/CNF inks with and without citric acid reveals the removal of ZnO upon sintering (Fig. 1F). Without citric acid, weak ZnO peaks appear in the printed electrodes at 31.8° and 34.4° , corresponding to the (100) and (002) planes, respectively. These peaks disappear with the addition of citric acids in the ink, indicating effective removal of surface oxides. Scanning electron microscopy (SEM) further shows that citric acid sintering promotes physical contacts between Zn particles, therefore enhancing the electrical conductivity (Fig. 1G). Overall, the conductivity of biodegradable inks can be tuned by optimizing particle size, loading content, and sintering agents. Compared with the conductivity of previously reported nondegradable 3D-printable conductive inks ranging from 0.1 to 10^5 S m^{-1} (fig. S8) (18, 33–36), the Mo/CNF, Zn/CNF, and Mo/PLCL inks demonstrate comparable conductivity. This can ensure desirable internal resistance for batteries, and suitable impedance for sensing, while offering the additional advantages of biocompatibility and biodegradability. Incorporating metallic particles with greater aspect ratios could further enhance conductivity by lowering the percolation threshold while maintaining printability (37).

Figure 1H shows the printing of the “THU” geometric structure using CNF-based inks, with a thickness of 3 mm (15 layers). A 10 mm-by-10 mm serpentine trace is also demonstrated using Mo/PLCL ink via 3D printing, achieving a print resolution of 160 μm . Moreover, a sandwich-structured Zn-Mo biodegradable battery is assembled layer by layer, comprising a Zn-negative electrode, an ALG/CMC gel electrolyte layer, and a Mo-positive electrode (Fig. 1H and fig. S9). The microstructure and morphology of the 3D-printed batteries are characterized by SEM. Top-view and cross-sectional SEM images of the Zn-Mo battery illustrate a printed filament width of approximately 400 μm and a layer thickness of around 200 μm (fig. S10).

The electrochemical characteristics of the 3D-printed Zn-Mo battery are investigated to evaluate its potential for delivering electrical cues to promote bone repair. Figure 1I shows the discharge behavior of Zn-Mo batteries with Zn electrodes fabricated from Zn/CNF inks with and without citric acid sintering agent at a discharge current density of $2 \mu\text{A cm}^{-2}$, using 3D-printed ALG/CMC gel electrolyte containing phosphate-buffered saline (PBS) solution. During battery discharge, the electrochemical reaction at the Zn electrode is expected to involve Zn oxidation ($\text{Zn} + 4\text{OH}^- \leftrightarrow \text{Zn}(\text{OH})_4^{2-} + 2e^- \leftrightarrow \text{ZnO} + 2\text{OH}^- + \text{H}_2\text{O} + 2e^-$), while oxygen reduction is anticipated at the Mo electrode ($\text{O}_2 + 2\text{H}^+ + 2e^- \leftrightarrow \text{H}_2\text{O}_2$, $\text{O}_2 + 2\text{H}_2\text{O} + 4e^- \leftrightarrow 4\text{OH}^-$) where OH^- and ROS could be generated (13, 16). The sintered Zn-Mo biodegradable battery exhibits an initial voltage of approximately 0.8 V and maintains over 0.6 V for more than 70 hours, with a total discharge lifespan extending to 100 hours. In contrast, the Zn-Mo battery with unsintered Zn electrodes shows an initial voltage of 0.6 V, which rapidly declines below 0.2 V after just 10 hours, indicating the importance of incorporating mild acids in the inks. Furthermore, a comparison of different sintering agents of Zn/CNF inks demonstrates that citric acid notably enhances the discharge performance (fig. S11A). A stable conductive network can be established in the electrodes with three printed layers (thickness, $\sim 0.6 \text{ mm}$), enabling the battery to consistently discharge at 0.8 V (fig. S11B). In addition, fig. S11C illustrates that the output voltage can be promoted

to 3.5 V by stacking four cells in series through 3D printing, at a discharge current density of $2 \mu\text{A cm}^{-2}$. As the discharge current density increases, the battery exhibits a reduction in output voltage and operational lifespan (fig. S11D). With a discharge current density of $15 \mu\text{A cm}^{-2}$, the battery offers a power density of $5.8 \mu\text{W cm}^{-2}$ (fig. S11E). Notably, the incorporation of 2-wt % citric acid into the electrolyte further extends the battery discharge time beyond 100 hours while maintaining a voltage consistently above 0.6 V (fig. S11F). To investigate the effect of electrode geometry on the electrochemical performance of the Zn-Mo galvanic battery, we also perform electrochemical impedance spectroscopy (EIS), open-circuit potential measurements, and galvanostatic discharge tests on batteries with varying electrode thicknesses (0.2 and 0.4 mm) and widths (0.4 and 1.0 mm, 1.5 mm) while keeping the length constant (10 mm) in agarose containing PBS. As shown in fig. S12A (Zn electrode impedance) and fig. S12B (Mo electrode impedance), increasing the electrode width leads to a decrease in charge transfer resistance component (the diameter of the semicircle of the EIS curve), which is attributed to the increased effective electrode surface area. In contrast, electrode thickness has little effect on the charge-transfer resistance, suggesting that thickness within the 0.2- to 0.4-mm range does not notably influence the electrode impedance. Moreover, ohmic series resistance component stays similar with different electrode geometry. The Zn-Mo battery is expected to operate under low-current conditions in vivo, as current is delivered through the surrounding tissue at the defect site, which presents a relatively high impedance. These results indicate that the impedance of the Zn and Mo electrodes does not cause significant voltage losses under the low operating currents relevant to our device. This performance is enabled by the porous electrode architecture, which provides a large electrochemically active surface area and thereby reduces interfacial resistance. Furthermore, the open-circuit potential measurements reveal that all tested groups maintain a stable open-circuit voltage (OCV) of approximately 0.8 to 0.9 V throughout the 4-hour monitoring period, with no significant difference observed among different thickness or width groups (fig. S12C). We further perform galvanostatic discharge tests at a current density of $50 \mu\text{A cm}^{-2}$. The results show little variation with electrode dimensions, indicating that moderate variations in electrode impedance do not notably affect the output voltage (fig. S12D).

When tested in a simulated physiological environment using 1 wt % agarose gel containing PBS, the Zn-Mo battery achieves a discharge time frame of approximately 90 hours at a current density of $50 \mu\text{A cm}^{-2}$ (Fig. 1J). The voltage fluctuations observed at approximately 60 hours during battery discharge (Fig. 1J) are likely due to gradual degradation of electrode components, and the dynamic process of the accumulation and redissolution of degradation byproducts, including ZnO and Zn(OH)₂, on the Zn electrode surface. These species may partially passivate the electrode, leading to intermittent increases in interfacial impedance and resulting in unstable voltage output during extended operation. In vivo, however, the device is expected to operate in a high-impedance regime, resulting in correspondingly low operating currents. Moderate variations in interfacial impedance are therefore not expected to substantially influence the electrochemical output. Under conditions simulating physiological conditions (agarose containing PBS), the OCV of the Zn-Mo galvanic cell is sustained for approximately 500 hours (~21 days) (fig. S13A), with a noticeable decline occurring after ~100 hours, likely because of degradation-induced increases in contact resistance. Overall, the electrical cues are expected to exert a positive

influence on bone regrowth, as early-stage electrical stimulation (7 to 20 days) has been reported to enhance long-term bone regeneration (13, 38). The underlying mechanism involves early electrical stimulation elevating intracellular Ca²⁺ dynamics and ROS, which in turn promote neovascularization, and direct stem cells through the up-regulation of key pathways such as TGF- β signaling toward osteogenic differentiation. These coordinated signals enhance osteogenic process, improve local microcirculation, and propel the early bone repair phase forward (6, 9, 39).

Since oxygen reduction is expected to occur at the Mo electrode during battery discharge, potentially modulating local ROS concentrations. ROS levels near the Mo electrodes are therefore assessed as ROS have been reported to enhance osteogenic differentiation of osteoblasts and stimulate the production of vascular endothelial growth factor (VEGF) in macrophages, thereby promoting angiogenesis during bone repair (8). The results show a significant increase in ROS production after 60 min of battery discharge (Fig. 1K). These findings suggest that Faradaic reactions of the battery could modulate the local microenvironment and facilitate bone regeneration. To evaluate potential pH changes induced by battery operation, we perform discharge tests in PBS at a current density of $50 \mu\text{A cm}^{-2}$. The results show that the pH near the electrode remains approximately 7.2 over 60 hours, indicating that the battery does not induce notable pH changes (fig. S13B).

Effects of Zn-Mo batteries on in vitro cell behavior

To evaluate the biocompatibility and osteogenic capacity of the Zn-Mo battery and its electrode materials, human bone mesenchymal stem cells (hBMSCs) are cultured with Zn (Zn group), Mo (Mo group), and Zn-Mo batteries (battery group). hBMSCs cultured on standard cell culture dishes serve as the control group (fig. S14). Metallic foils are used in in vitro studies to enable stable immobilization, facilitating evaluation of the effects of the electric field and mimicking the in vivo deployment of the self-electrified scaffold. Owing to the notably enhanced effective surface area of the printed electrodes, their electrode impedance (mainly charge-transfer resistance) is lower than that of the foil electrodes (fig. S15A), indicating that the printed electrodes do not impede battery performance. Galvanostatic discharge tests in agarose gel at a current density of 2 and $10 \mu\text{A cm}^{-2}$ show that the discharge voltage of the foil battery is slightly lower than that of the 3D-printed battery, primarily because of the larger specific surface area of the 3D-printed electrodes (fig. S15B). These results demonstrate that printed batteries can deliver electrochemical performance comparable to, or slightly better than, that of foil batteries. Therefore, in vitro studies using foil electrodes can be used to reflect the osteogenic effects of the printed battery. During in vitro experiments, the Zn and Mo electrodes encapsulated by PLGA (poly(lactic-co-glycolic acid)) are placed in standard six-well plates (electrode spacing, ~35 mm). The OCV is assessed by measuring the voltage of Zn-Mo electrodes immersed in cell culture medium, as shown in fig. S15C. The battery maintains an OCV of approximately 0.65 V in cell culture medium for over 4 days. Galvanostatic discharge measurements at a current density of $10 \mu\text{A cm}^{-2}$ in PBS show that the PLGA coating does not impede battery discharge (fig. S15D). Calcein-AM/propidium iodide (PI) staining is used to assess cell viability after 24 hours. Vital cells (green) predominate across all groups, with only a small proportion of PI-stained apoptotic cells (red) observed (Fig. 2A). Quantitative analysis of cell viability using ImageJ (Fig. 2B) reveals no significant differences in

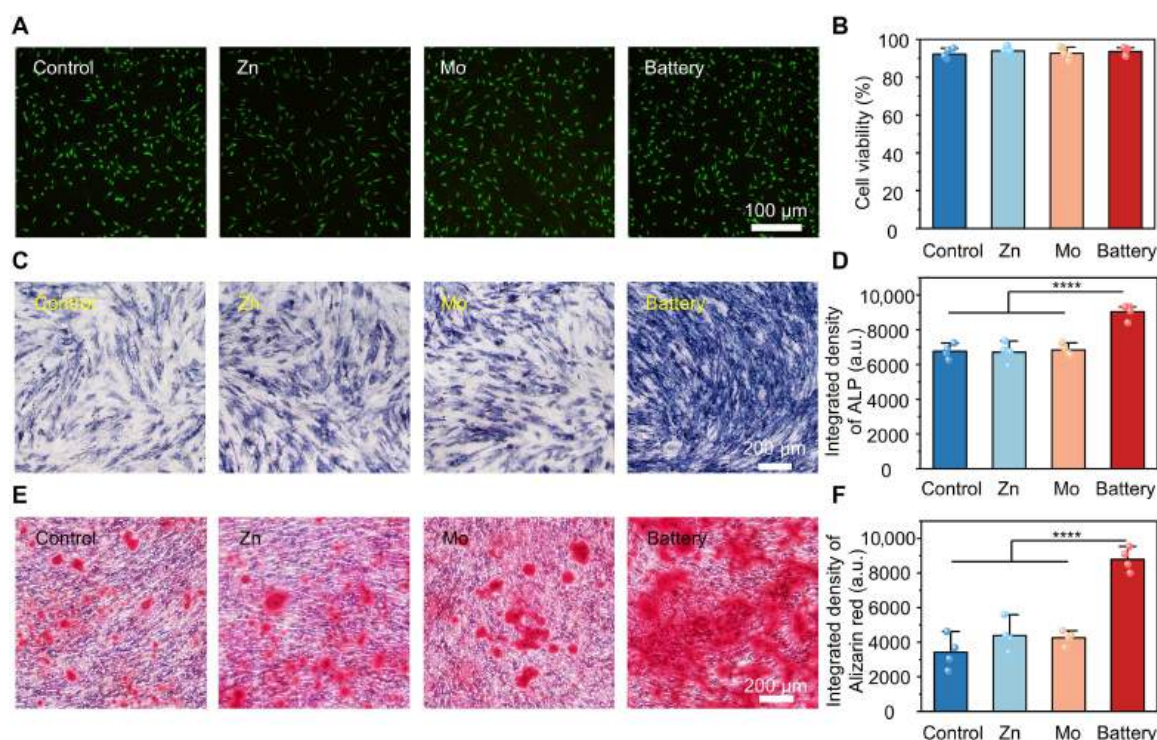


Fig. 2. Influence of Zn-Mo batteries on in vitro biocompatibility and osteogenic capacity. (A) Live/dead staining of hBMSCs cultured with Zn-Mo battery, Zn electrodes, and Mo electrodes. Cells cultured in standard cell culture dishes serves as the control group. (B) Cell viability of hBMSCs at 24 hours. (C) ALP staining of hBMSCs on day 7. (D) Quantitative evaluation of the ALP staining integrated density on day 7. (E) ARS of hBMSCs on day 14. (F) Quantitative evaluation of the ARS staining integrated density on day 14. In (B), (D), and (F), data are means $\pm$ standard deviation. One-way ANOVA with Tukey's multiple comparisons test, **** $P < 0.0001$. In (A), $n = 5$ independent experiments; in (D) and (F) $n = 4$ independent experiments.

cell viability among the control, Zn, Mo, and battery groups, confirming excellent biocompatibility of the battery. Phalloidin/4',6-diamidino-2-phenylindole (DAPI) fluorescent staining represents an elongated, spindle-like cellular morphology during coculture with the battery (fig. S16). To further investigate the osteogenic inductive effect of electrical cues, alkaline phosphatase (ALP) staining and quantitative analysis are performed on day 7, followed by Alizarin Red S (ARS) staining and quantification on day 14. The battery group exhibits significantly enhanced ALP activity and calcium nodule deposition compared to the other groups, including those cultured with Zn or Mo electrodes (Fig. 2, C and F), and those treated with the corresponding ionic leachates including Zn, Mo, and Mix (Zn + Mo) (fig. S17). These results indicate that the enhanced osteogenic differentiation of hBMSCs is induced by the active electrical cues from the Zn-Mo batteries rather than by ionic effects from the electrode materials. We perform inductively coupled plasma optical emission spectroscopy to quantify Zn and Mo ion concentrations in the culture media supernatants in the presence of Zn-Mo batteries (fig. S18). The culture medium was replaced every 2 days, with half of the medium refreshed at each change. The detected levels fall within the biosafety and osteogenic ranges for hBMSCs and bone regeneration, as reported for Zn (0.01 to 0.08 mM) and Mo (0.02 to 0.32 mM) (40–42). Consistently, no significant cytotoxicity or morphological abnormalities are observed in hBMSCs cocultured with the Zn-Mo battery.

Electrical cues have been shown to activate voltage-gated calcium channels, triggering extracellular Ca^{2+} influx that plays a pivotal

role in regulating hBMSCs differentiation, neovascularization, cell proliferation, and the secretion of proregenerative growth factors, ultimately facilitating bone tissue regeneration (6, 9, 43). Effects of Zn-Mo batteries on intracellular calcium levels are therefore assessed using the calcium-sensitive probe Fluo-4 AM. hBMSCs exhibit a progressive increase in green fluorescence following the introduction of the Zn-Mo cell in the culture dish (Fig. 3A), with fluorescence intensity peaking at 60 min (Fig. 3B), indicating elevated Ca^{2+} activity. The battery group shows significantly higher levels of fluorescence intensity than other groups at 60 min (Fig. 3, C and D), suggesting that the electric field generated by the battery enhances calcium dynamics.

Moreover, although excessive ROS have been regarded as pathological agents that induce apoptosis, physiologic levels of ROS are increasingly recognized as key mediators of electrostimulation-enhanced bone regeneration (9, 44). Moderate ROS generation in response to electrical stimulation has been reported to up-regulate osteogenic gene expression, enhance cell differentiation, and promote VEGF-mediated angiogenesis (9). We therefore also investigate ROS levels in the presence of the Zn-Mo battery. ROS levels measured 3 hours and 1 day after battery introduction are elevated compared with the control group (fig. S19). To further validate the pro-osteogenic roles of elevated ROS and Ca^{2+} levels introduced by the Zn-Mo battery, we conduct supplementary experiments using the ROS scavenger *N*-acetylcysteine (NAC) (45–47) and the calcium channel blocker nifedipine (NIF) (48, 49) as intervention agents. The Zn-Mo battery-induced enhancement of osteogenic differentiation

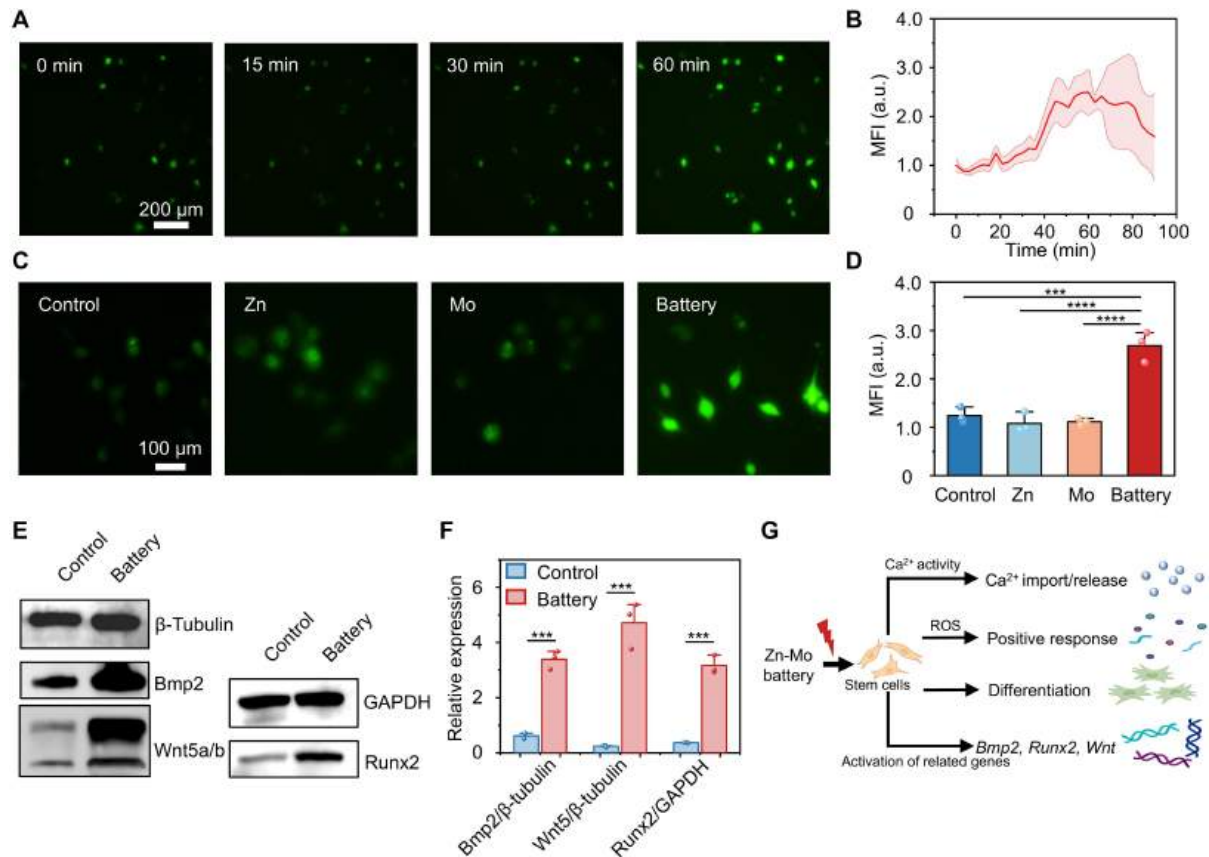


Fig. 3. Influence of Zn-Mo battery on intracellular calcium levels and osteogenic markers of hBMSCs. (A) Fluorescent images indicating the intracellular calcium level of hBMSCs over time in the presence of Zn-Mo battery detected by Fluo-4 staining (calcium, green). (B) Mean fluorescence intensity (MFI) of hBMSCs over time in the battery group. (C) Fluorescent images indicating the intracellular calcium level of hBMSCs after coculturing for 60 min in the control, Zn, Mo, and battery groups, detected by Fluo-4 staining (calcium, green). (D) MFI of hBMSCs after coculturing for 60 min in the control, Zn, Mo, and battery groups. (E) Western blot analysis of the expression of essential proteins involved in osteogenic signaling pathways in the control group and battery groups. (F) Relative expression of Bmp2, Wnt5, and Runx2. (G) Potential positive effects of Zn-Mo battery on osteogenic differentiation. Created in BioRender. Hua, W. (2026) <https://BioRender.com/gpmu07x>. In (D), one-way ANOVA with Tukey's multiple comparisons test were used for the statistical analysis. In (F), Student's *t* test was used for the statistical analysis. ****P* < 0.001, and *****P* < 0.0001. In (A), (C), and (E), *n* = 3 independent experiments.

is significantly attenuated by the addition of 500 μM NAC or 50 μM NIF (fig. S20), indicating that its pro-osteogenic effects are associated with elevated intracellular ROS and Ca^{2+} levels.

Furthermore, Western blot assay reveals substantial up-regulation of osteogenic markers, including BMP2, WNT5A/5B, and RUNX2 in hBMSCs cultured with Zn-Mo batteries for 14 days (Fig. 3, E and F). A schematic summarizing the potential beneficial effects of the Zn-Mo battery is shown in Fig. 3G: The delivered electrical cues enhance Ca^{2+} influx and ROS generation via Faradaic reactions and activate associated signaling pathways, ultimately up-regulating osteogenic genes and promoting bone regeneration.

Evaluation of bone regeneration by the biodegradable and self-electrified scaffold

To assess the bone-regenerative performance of the self-electrified scaffold, scaffolds with Zn-Mo batteries (battery group) (Fig. 4, A and B), scaffolds with single Mo or Zn electrodes (Mo or Zn group), and MC/PCL scaffolds (control group) are implanted into critical-sized cranial defects (~5 mm in diameter) in Sprague-Dawley (SD) rats (Fig. 4C). Micro-CT quantitation reveals extensive bone regeneration

in the battery group compared with the other groups (Fig. 4D and fig. S21). Specifically, at 12 weeks postimplantation, the battery group exhibits extensive trabecular bone structure within the defect, whereas the other groups show much less bone formation at the injury site, indicating delayed repair (Fig. 4D). The implanted self-electrified scaffold is visible in 3D reconstruction images at week 0 (Fig. 4E, left), and enhanced bone regeneration is confirmed at 12 weeks compared to the contralateral control side (Fig. 4E, right). Quantitative CT analysis (Fig. 4F) shows a significant increase in bone thickness in the battery group (2.27 ± 0.23 mm), compared to the control (1.46 ± 0.22 mm), Mo (1.74 ± 0.16 mm), and Zn (1.47 ± 0.21 mm) groups (*P* < 0.0001 versus control; *P* < 0.05 versus Mo and Zn) at 12 weeks postoperatively. Bone mineral density (BMD) and bone volume/total volume (BV/TV) show similar overall trends, whereas BMD exhibits more pronounced differences among the groups (Fig. 4G and fig. S22A). BMD are significantly higher in the battery group (0.36 ± 0.02 g mm^{-3}) compared to the control (0.24 ± 0.02 g mm^{-3}), Mo (0.26 ± 0.02 g mm^{-3}), and Zn (0.23 ± 0.01 g mm^{-3}) groups at 12 weeks postimplantation (*P* < 0.0001; Fig. 4G). The apparent plateau in BV/TV and BMD values in the battery group from week 2 to

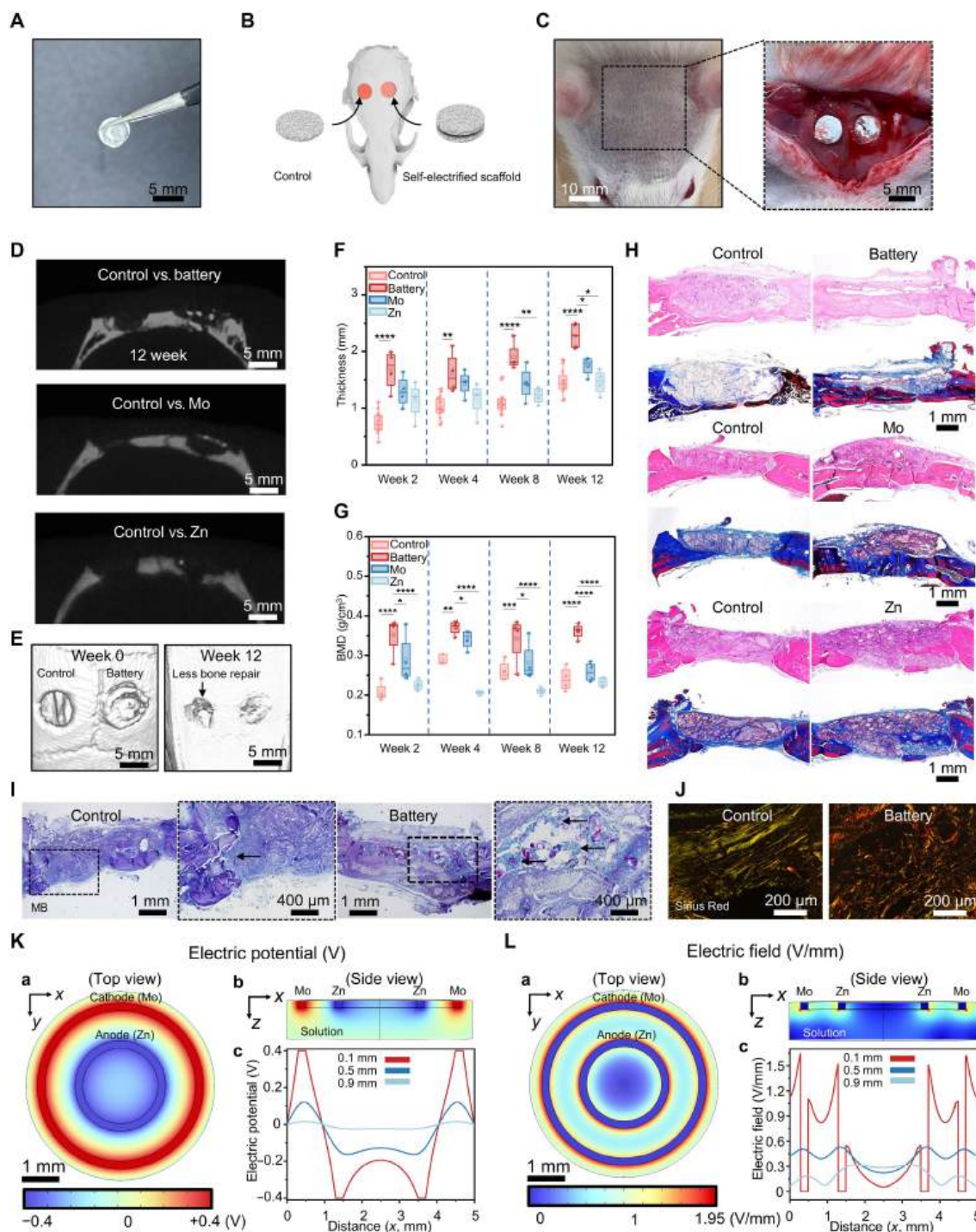


Fig. 4. Evaluation of bone regeneration by the biodegradable and self-electrified scaffold. (A) Photograph of the battery-based self-electrified scaffold. (B) Schematic illustration of the battery-based self-electrified scaffold. (C) Photographs of critical-sized cranial defects (~5 mm in diameter) created in a rat model and implanted scaffolds to facilitate bone regeneration. (D) Radiographic evaluation of cranial defect repair at 12 weeks postimplantation based on micro-CT imaging. (E) 3D reconstruction images of bone samples collected at 0 week and 12 weeks after implantation. (F) Statistical results of thickness for newly formed bone tissue. (G) Statistical results of BMD for newly formed bone tissue. (H) H&E and Masson's trichrome staining images of regenerated bone tissues with various scaffolds 12 weeks postimplantation. (I) MB staining images of regenerated bone tissues 4 weeks postimplantation. (J) Sirius Red staining images of regenerated bone tissues 12 weeks postimplantation. (K) and (L), FEA. Electric potential [(K (a)), [K (b)]] and electrical field [(L (a)), [L (b)]] distributions between the two circular electrodes with a Zn-Mo battery output voltage of 0.8 V in solution mimicking cranial tissues. [K (c)] and [L (c)] showing the distribution profile at different longitudinal positions from the scaffold surface ($Z = 0.1, 0.5$, and 0.9 mm). In (F) and (G), the box plot presents the median (center line), lower quartile (lower border), upper quartile (upper border), maximum (upper whisker), and minimum (lower whisker), which are ≤ 1.5 times the interquartile range. One-way ANOVA with Tukey's multiple comparisons test were used for the statistical analysis. $*P < 0.05$, $**P < 0.01$, $***P < 0.001$ and $****P < 0.0001$. In (D) to (J), $n = 4$ independent experiments.

week 12 is likely influenced by artifacts arising from metallic electrode components that introduce high-density signals, particularly at early time points. Accordingly, all groups containing metal electrodes show higher BMD than the control group at week 0 (fig. S22B). As the metals degrade, BMD values in the Zn and Mo groups decrease. In contrast, BMD values in the battery group remain largely unchanged, likely because accelerated bone formation compensates for the reduced contrast associated with metal degradation. As the device degrades and is nearly completely resorbed by week 12, the measured BV/TV and BMD values more accurately represent true bone mineralization, as further supported by histological evidence (Fig. 4H).

Hematoxylin and eosin (H&E) and Masson's trichrome staining of frontal sections at 12 weeks postimplantation reveals extensive, well-organized bone formation in the battery group, with new bone bridging the defect and exhibiting greater maturation than in the other groups (Fig. 4H). Specifically, Masson's staining shows a predominance of mature bone matrix (red area) in the battery group, and new bone tissue have largely covered the entire defect area. By contrast, most control groups exhibit limited the formation of newly formed bone, characterized by immature (blue area) and disorganized structure, along with extensive fibrous connective tissue. The Mo and Zn groups show moderate new bone formation, primarily characterized by insufficient mineralization. Time-course histological analyses of transverse section at 4 and 12 weeks further confirms accelerated bone regeneration and maturation in the battery group (fig. S23). In addition, only a small number of undegraded metal particles are observed in the H&E staining at 12 weeks, indicating that the battery components have nearly fully biodegraded. These results indicate that the self-electrified scaffold promotes both the regeneration and maturation of cranial bone defects. During the early stages of bone regeneration, the synthesis and secretion of collagen remodel the ECM, thereby facilitating cell migration, osteogenic differentiation and subsequent mineralization of HA (50). To assess collagen deposition and remodeling, methylene blue (MB) and Sirius Red stainings are performed. At 4 weeks postimplantation, MB staining reveals extensive collagen deposition in the battery group, characterized by broad, densely stained azure regions, in contrast to the sparse, dotted collagen observed in the control group (Fig. 4I). Sirius Red staining at 12 weeks demonstrates a predominance of type I collagen (red) in the battery group, whereas the control group displays mainly type III collagen (green). Mature type I collagen is visualized in red and yellow, while green indicates immature collagen. The results of Sirius Red staining indicate that the self-electrified scaffold promotes collagen maturation at an earlier stage. In addition, immunofluorescence staining at 4 weeks postoperatively reveals elevated expression of osteogenic markers, including *Bmp2*, *Ocn*, and *Bsp*, in the battery group (fig. S24), which are involved in the process of collagen mineralization. Moreover, it is worth mentioning that unlike classical direct-current electrical stimulation, where anodic reactions can cause potential harmful effects such as acidification or toxic electrode dissolution (5), our battery generates electrical stimulation through a galvanic Zn-Mo reaction that produces biocompatible species at the anode site, primarily Zn^{2+} at safe concentrations. The resulting electric field and current promote osteogenic signaling by enhancing Ca^{2+} influx, ROS generation, and downstream gene expression. Consistent with these mechanisms, the battery significantly improves bone regeneration in vivo, whereas Zn or Mo electrodes alone show no appreciable osteogenic benefit, demonstrating

that the integrated anode-cathode system is both safe and therapeutically effective.

Furthermore, we perform finite element analysis (FEA) to systematically quantify the spatial distribution and effective penetration of the electric field within the cranial defect region. We establish a 3D FEA model that accurately reflects the geometric dimensions and structural parameters of the printed Zn-Mo battery, including the ring-shaped Zn anode and Mo cathode. Analysis of the electric potential distribution shows that the potential spans a millimeter-scale range, covering the entire 5-mm-diameter scaffold (Fig. 4K). Along the *z* axis, 0.5 mm from the electrode surface, the potential ($>\pm 0.1$ V) retains more than 25% of its initial surface value. Moreover, the electric field exhibits a clear radial gradient between the Zn anode and Mo cathode, attenuating from the electrode surface into the surrounding tissue while remaining above 0.8 V mm^{-1} between the electrodes (Fig. 4L). Along the *z* axis, the field remains approximately 0.2 V mm^{-1} at 0.5 and 0.9 mm from the electrode surface, demonstrating a desirable vertical penetration depth comparable to the typical thickness of rodent cranial bone (~ 1 mm). The lateral and vertical electric field strengths fall within the range reported in the literature (0.025 to 1 V mm^{-1}) that effectively regulate osteogenesis-related cellular behaviors, including osteoblast and mesenchymal stem cell migration, osteogenic factor secretion, cytoskeletal remodeling, and proliferation (6, 7, 9, 10, 51–53). FEA is further used to investigate the effects of electrode dimensions on the electric field distribution. Parametric sweeps of electrode thickness (*t*), width (*w*), and spacing (*s*) reveal that increasing *t* enhances fringing effects and improves field penetration (fig. S25A), increasing *w* broadens the spatial coverage of effective stimulation (fig. S25B), and decreasing *s* yields stronger localized field intensity following $E = V/d$ (fig. S25C). These simulation results support that the Zn-Mo battery-generated electric field effectively penetrates surrounding cranial tissue and sustains a desirable stimulation intensity, thereby promoting bone repair.

To further investigate the mechanisms underlying the bone regenerative effects of the self-electrified scaffold, transcriptome sequencing is performed on the control and battery groups at the 3 weeks postimplantation. Analysis of the volcano plot and heatmap reveal a total of 693 differentially expressed genes (DEGs) between the control and battery groups, including 183 up-regulated and 510 down-regulated genes (Fig. 5, A and B). Heatmaps indicate the up-regulation of osteogenesis-related (e.g., *Bmpr1b*, *Ibsp*, and *Adam8*; Fig. 5C), angiogenesis-related (e.g., *Hif1a*, *Mmp2*, and *Angptl4*; Fig. 5D), and ECM-related genes (e.g., *Col1a1*, *Col1a2*, and *Col5a3*; Fig. 5E) in the battery group. Moreover, Gene Ontology (GO) analysis (Fig. 5F) demonstrates that these DEGs are significantly enriched in pertinent biological processes (e.g., ossification and tissue remodeling), cellular components (e.g., ECM and collagen trimer), and molecular functions (e.g., collagen binding and cytokine activity). Kyoto Encyclopedia of Genes and Genomes (KEGG) pathway analysis is further performed on DEGs to reveal the underlying osteogenic signaling pathways in which they are enriched. Among these pathways, the ECM-receptor interaction and TGF- β signaling pathway, both critical for ECM organization and osteogenesis, are notably up-regulated by electrical stimulation (fig. S26). This up-regulation may be associated with the increase in intracellular Ca^{2+} ions and ROS induced by electrical cues. Ca^{2+} ions act as pivotal mediators of cell-ECM communication and as second messengers that broadly cross-talk with TGF- β signaling to cooperatively direct stem cell fate (6, 54, 55). Meanwhile, ROS have been shown to influence the TGF- β pathway,

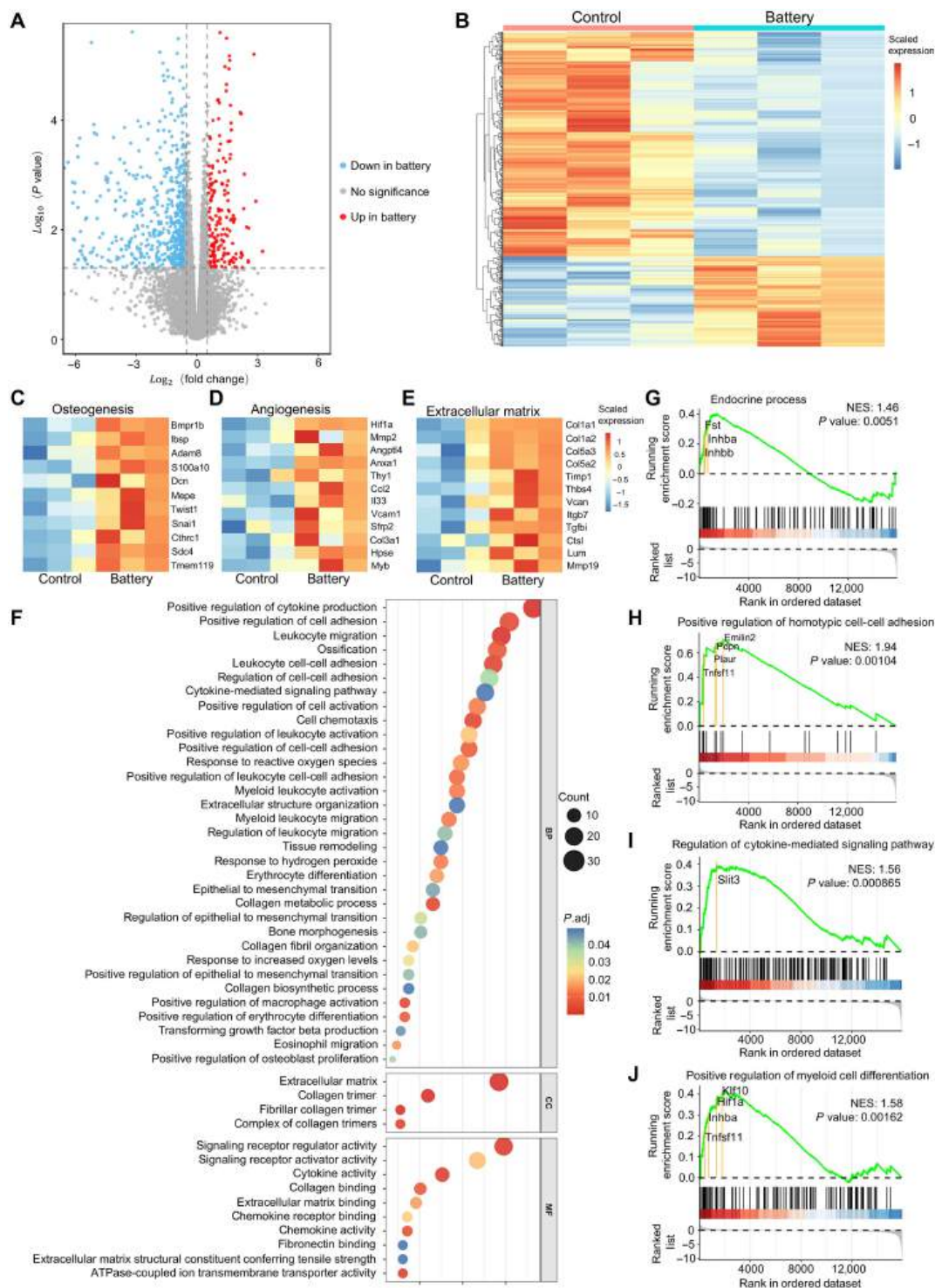


Fig. 5. Transcriptomic profiling and functional enrichment analysis. (A) Volcano plot of DEGs (red: up-regulated; blue: down-regulated). (B) Heatmap of all identified DEGs. Heatmaps showing expression of genes related to (C) osteogenesis, (D) angiogenesis, and (E) ECM. (F) GO enrichment analysis across BP (biological process), CC (cellular component), and MF (molecular function) categories. (G to J) GSEA of selected signaling pathways and biological processes. $n = 3$ independent experiments.

promoting both angiogenesis and stem cell differentiation (8). Gene set enrichment analysis (GSEA) confirms that the TGF- β signaling pathway related genes (e.g., *Inhbb*, *Fst*, and *Inhba*) are activated in the battery group (Fig. 5G). GSEA further indicates that electrical stimulation enriches key pathways supporting bone repair, including the positive regulation of cell-cell adhesion, cytokine-mediated signaling, and myeloid cell differentiation (Fig. 5, H to J). This suggests that electrical cues promote bone regeneration by enhancing tissue cohesion, modulating inflammatory responses, and driving osteoclast-related remodeling. Collectively, these transcriptional profiles demonstrate that electrical cues from the scaffold activate coordinated signaling networks that promote bone regeneration through matrix production, osteogenic differentiation, and adaptive remodeling. Nevertheless, although the transcriptional profiles from the animal model provide some evidence linking Ca^{2+} and ROS signaling, they do not fully replicate the in vitro results. This is likely due to the highly complex in vivo environment, in which electrically induced signaling pathways are temporally dynamic and involve extensive multicellular cross-talk. Elucidating deeper underlying in vivo mechanisms remains an active area of investigation and a key direction for future work.

Multifunctional cranial device for bone defect treatment and monitoring

Early-stage processes such as cell proliferation, matrix deposition, and the initiation of mineralization are critical for successful bone regeneration (56). Assessment during this stage is therefore essential for timely diagnosis and optimal therapeutic outcomes (57, 58). However, conventional diagnostic modalities such as x-ray imaging rely on detectable mineralization and are insufficient for identifying early deficiencies in repair (59, 60). Alternatively, bioimpedance, closely linked to the extracellular composition that dynamically evolves during early regeneration, could offer a promising diagnostic approach (61–65). Despite this potential, the development of biodegradable, integrated devices capable of both promoting bone regeneration and enabling continuous monitoring remains a challenge. By integrating a self-electrified scaffold with flexible Mo/PLCL electrodes, we develop a multifunctional cranial device capable of delivering electrical cues to the injury site to promote tissue regrowth while simultaneously monitoring bone healing via impedance measurements (Fig. 6, A to C).

We first evaluated the in vitro electrical properties of flexible, biodegradable Mo/PLCL electrodes fabricated via 3D printing. These electrodes exhibit stable performance upon immersion in PBS at 37°C for up to 4 weeks, with average impedance at 10 kHz increasing from $17.4 \pm 0.5 \text{ k}\Omega$ on day 0 to $46.6 \pm 2.1 \text{ k}\Omega$ on day 28, followed by a marked rise after day 32 due to materials degradation (Fig. 6D). The desirable operational timeframe over 4 weeks is likely attributable to the stability of PLCL matrix and its minimal swelling rate. During the early stages of bone regeneration, collagen deposition and subsequent mineralization into HA within the ECM are key processes underlying successful bone repair. To assess the associated bioimpedance changes, we incorporate varying concentrations of collagen or HA into a 2-wt % sodium ALG hydrogel matrix to simulate compositional evolution during regeneration. Impedance within the 1- to 32-kHz frequency range has been reported to correlate with ECM resistance, which reflects the status of bone healing (62, 66). Accordingly, impedance at 10 kHz was selected to assess tissue compositional changes under both simulated and in vivo conditions. EIS measurements using Mo/PLCL electrodes demonstrates that increases

in collagen or HA content are accompanied by corresponding increases in impedance (Fig. 6, E and F, and fig. S27). Specifically, impedance at 10 kHz increases notably with increased collagen concentration, rising from $29.6 \pm 1.4 \text{ k}\Omega$ at 20 wt % to $68.2 \pm 3.8 \text{ k}\Omega$ at 25 wt % (Fig. 6E). HA incorporation also leads to a notable increase in impedance with rising HA content (Fig. 6F). These results suggest that bioimpedance could serve as a sensitive indicator of ECM composition changes and, thus, provide a robust tool to monitor bone healing progression at the early stage.

The biodegradable, multifunctional device, integrating a self-electrified scaffold with bioimpedance electrodes, is implanted into rat cranial defects (5 mm in diameter) to evaluate in vivo performance (Fig. 6C). Mo/PLCL electrodes without the self-electrified scaffold serves as the control group. Impedance measurements over time and corresponding fitting curves using equivalent circuits are shown in Fig. 6, G and H. Impedance at 10 kHz remains nearly unchanged in both groups at 1 week postimplantation. From weeks 2 to 4, however, impedance progressively increased in the multifunctional cranial device group, likely reflecting fibrous tissue formation, collagen deposition, and mineralization during early bone regeneration. In contrast, impedance in the control group remains low throughout the 4-week period, indicating minimal compositional change at the defect site. EIS spectra are fitted using an equivalent circuit model for bone healing as reported previously (67). In this model, C_p represents the capacitance between electrodes and flexible printed circuit (FPC) lines; CPE_{dl} denotes the constant phase element at the electrode-tissue interface; C_t and R_t correspond to cell membrane capacitance and intracellular resistance, respectively; and R_2 reflects ECM resistance, which correlates with compositional changes of regenerated tissues. Notably, impedance at 10 kHz features a marked rise in R_2 , from $2.5 \pm 0.1 \text{ k}\Omega$ at week 0 to $85.6 \pm 2.3 \text{ k}\Omega$ at week 2, reaching $514.1 \pm 142.4 \text{ k}\Omega$ by week 4 in the cranial device group. In comparison, R_2 in the control group increases only modestly from $3.4 \pm 0.3 \text{ k}\Omega$ to $9.9 \pm 4.7 \text{ k}\Omega$ over the same period. The divergent impedance trajectories between the cranial device and control groups reflect distinct alterations in ECM composition. The marked increase in impedance observed in the cranial device group suggests enhanced collagen deposition and matrix mineralization, indicating improved healing progression likely driven by the self-electrified scaffold.

Moreover, we conduct additional experiments to evaluate other physiological factors that may contribute to impedance changes. We first examine the effects of tissue hydration using sodium ALG hydrogels with varying water contents to mimic different hydration states. Although impedance at 10 kHz increases as water content decreases (fig. S28A), the measured values (10 to 100 k Ω) remain notably lower than those observed during enhanced bone healing at week 4 (~1000 k Ω). We then assess the potential influence of inflammation, which often results in excess interstitial fluid. Using blood to simulate such conditions, we find that the impedance at 10 kHz remains relatively low (~10 k Ω) (fig. S28B). To mimic possible fibrosis at the electrode-tissue interface, we measure impedance through skin tissues, which yields values of ~20 k Ω at 10 kHz (fig. S28B). Overall, these factors do not produce impedance increases comparable to those associated with improved bone healing at week 4 when collagen deposition and matrix mineralization are pronounced. Nonetheless, other untested factors may contribute to bioimpedance and warrant further investigation.

Histological analysis at 4 weeks postimplantation corroborates these findings (Fig. 6J). H&E staining reveals substantial tissue regeneration

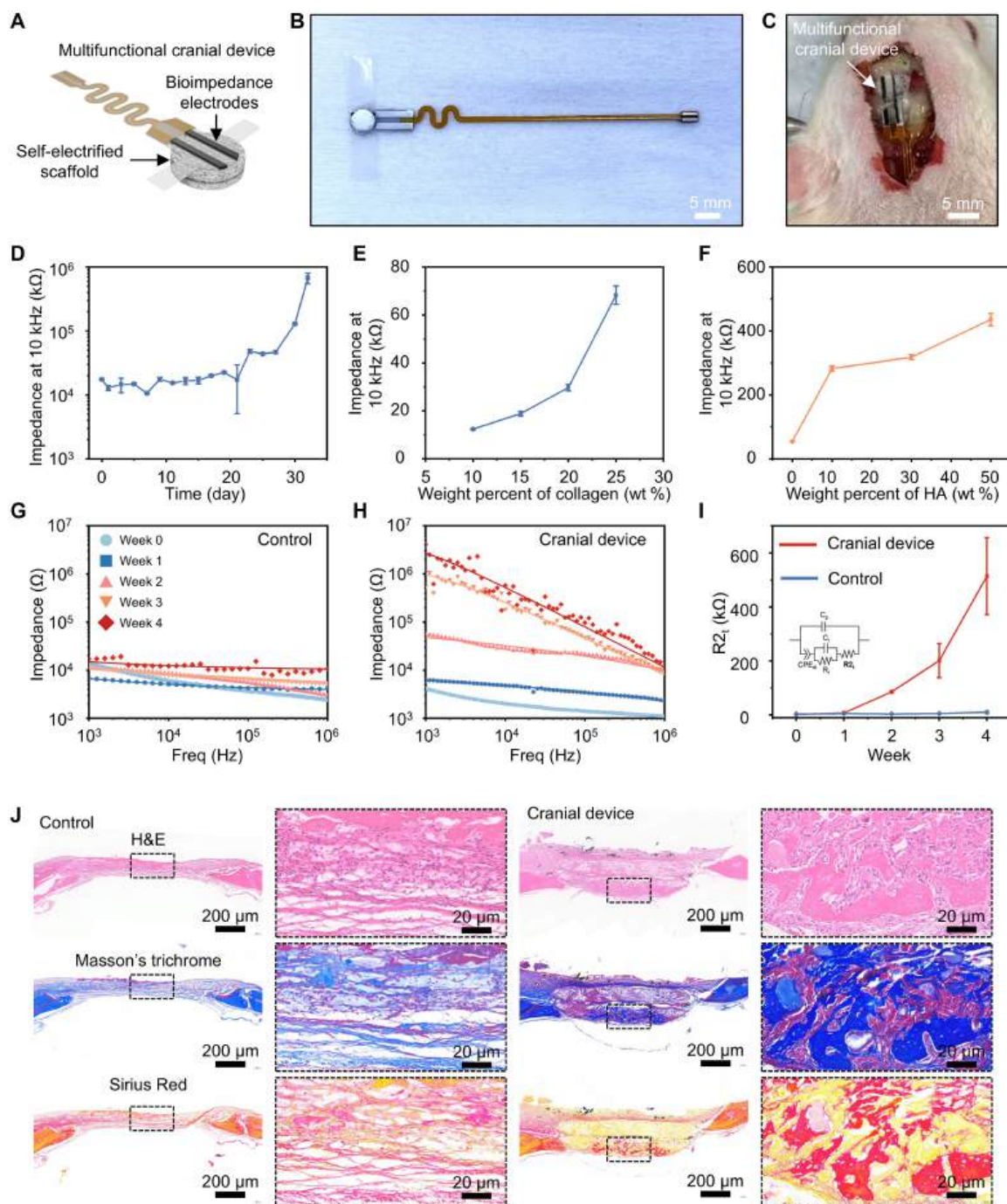


Fig. 6. In vivo evaluation of the biodegradable and multifunctional cranial device. (A) Schematic illustration of the multifunctional cranial device. (B) Photograph of the cranial device. (C) Surgical images of the implanted cranial device. (D) Change of impedance at 10 kHz over time of Mo/PLCL electrodes immersed in PBS at 37°C. (E) Impedance at 10 kHz of simulated ECM with varying concentrations of collagen. (F) Impedance at 10 kHz of simulated ECM with varying concentrations of HA. (G) EIS measurement at the cranial defect over a 4-week period of the control group. (H) EIS measurement at the cranial defect over a 4-week period of the cranial device group. (I) Summarized R_{2t} at weeks 0, 1, 2, 3, and 4 of the control and cranial device group. Impedance data are fitted using an equivalent circuit model (inset). (J) H&E Masson's trichrome and Sirius Red staining images of regenerated bone tissues at 4 weeks postimplantation. In (G), (H), and (J), $n = 3$ Independent experiments.

in the cranial device group, with newly formed bone and fibrous tissue evident within the defect region, while the control group shows only sparse fibrous tissue and negligible bone formation. Masson's trichrome staining further confirms minimal bone formation in the control group, in contrast to abundant, blue-stained new bone tissue, fibrous matrix, and neovascularization in the cranial device group. A thin layer of bone plate is also observed at the upper defect boundary in the cranial device group. Sirius Red staining demonstrates extensive red-stained collagen deposition and numerous yellow-stained cells of the connective tissue in the cranial device group, indicating enhanced collagen and connective tissue formation. Together, impedance measurements and histological analyses suggest that the multifunctional cranial device not only promotes bone regeneration but also enables continuous monitoring of healing progression through bioimpedance sensing.

As shown in Fig. 6D, the Mo/PLCL electrode maintains stable impedance for approximately 4 weeks. Accordingly, we monitored the impedance of cranial defects over this period. Longer-term monitoring may be enabled through optimized encapsulation layers or the incorporation of sintering agents for Mo particles to extend operational life time. Notably, although bone regeneration typically extends to 12 weeks, early-stage monitoring can still provide meaningful prognostic information (68–71) as early microenvironmental cues and cell fate decisions lay the foundation for critical callus formation and vascularization, which are closely correlated with the extent and quality of subsequent bone regeneration and remodeling (2, 72), enabling prediction of defect healing and discrimination between treatments with rapid versus delayed osteogenic and mineralization responses.

Furthermore, to evaluate the degradation behavior of the multifunctional cranial device, its constituent components are subjected to accelerated dissolution testing in PBS at 60°C (fig. S29, A and B). The self-electrified scaffold undergoes gradual degradation over 7 days, with its main structure fully disintegrated, leaving behind fragmented electrode residues and MC/PCL particulates, which requires a longer period to fully degrade. The Mo/PLCL electrodes exhibit progressive degradation via hydrolysis of the PLCL matrix and dissolution of the Mo components. Most of the materials degrades by day 12, with complete degradation of residual polymer fragments expected to occur over a longer timescale. To further assess biocompatibility and *in vivo* biodegradation, the self-electrified scaffold and Mo/PLCL electrodes are implanted subcutaneously in SD rats. As shown in fig. S29 (C and D), both components gradually degrade over 8 weeks in response to physiological fluids and enzymatic activity. By week 12, histological examination of major organs, including the lungs, liver, spleen, kidneys, brain, and heart, reveals no pathological abnormalities (fig. S29E), suggesting the favorable biocompatibility of the multifunctional cranial device.

In all, early monitoring of bone healing is critical for optimizing clinical treatment as it provides real-time insights into the dynamic processes at the injury site. For instance, timely feedback enables clinicians to distinguish normal healing from nonunion or impaired healing induced by infection, thereby allowing prompt interventions such as adjustments to physical stimulation regimens or the timely administration of biological therapies (e.g., growth factors and antibiotics) (67, 73, 74). As healing trajectories vary across different individuals, bioimpedance measurements could enable a means of personalizing treatment to both optimize the healing environment and minimize long-term complications, ultimately improving patient outcomes.

DISCUSSION

In this study, we develop a biodegradable and multifunctional cranial device that concurrently promotes bone regeneration and enables bioimpedance monitoring during the critical early stages of healing. Fabricated via 3D printing with biodegradable conductive inks, the device allows rapid, customizable production. Its self-electrified design autonomously delivers localized electrical cues without external interventions. In a rodent cranial defect model, this self-electrified device autonomously delivers localized electrical stimulation that triggers intracellular calcium dynamics and ROS production, activates osteogenic pathways such as TGF- β signaling, and up-regulates osteogenesis-related genes, collectively accelerating bone regeneration. This aligns with established mechanisms of bioelectronic and electroactive implants (9, 73, 75). Recent advances in bioelectronic approaches to bone regeneration include motion-driven triboelectric or piezoelectric systems that accelerate bone repair via electrical pulses (7, 10), magnetoelectric membranes that enhance surface polarization under external magnetic fields to promote cranial defect repair (12), and optoelectronic scaffolds incorporating silicon-based components that wirelessly stimulate osteogenic differentiation and osteoconductivity under near-infrared illumination (13). In contrast to these platforms, which typically rely on nondegradable materials or external energy inputs (e.g., mechanical motion or light), the self-electrified device reported here operates without an auxiliary power source and is biodegradable, obviating the need for secondary removal procedures. Notably, this platform further extends current paradigms by integrating real-time diagnostic feedback through bioimpedance monitoring, a capability absent from existing therapeutic-only implants. Future advances in ink formulation may enable higher-resolution printing for more complex circuitry, while integration with wireless modules could allow untethered monitoring. Moreover, future iterations of the device could incorporate biodegradable connectors to external wiring, as demonstrated in prior studies (76), thereby eliminating the need for follow-up surgeries to remove the nondegradable interconnects and minimizing associated risks. Coupling additional biophysical or biochemical signals may provide more comprehensive insights into the assessment of healing. As bone healing varies across age groups and disease conditions (e.g., diabetes and osteoporosis) (77–79), future studies could use diverse animal models to evaluate the therapeutic efficacy of the cranial device under a broader range of physiological conditions. Overall, this work presents a biodegradable, self-electrified device that integrates therapeutic stimulation with real-time diagnostics, offering a promising strategy for intelligent regenerative systems.

MATERIALS AND METHODS

Preparation of biodegradable inks for 3D printing CNF-based inks

CNFs (diameter: 4 to 10 nm, length: 1 to 3 μ m; Macklin Biochemical Technology Co. Ltd., China) were initially dispersed in deionized water using a vortex mixer at a weight-to-volume (w/v) ratio of 5:100. The resulting dispersion was stored overnight at 4°C. To prepare the Zn ink, Zn powders (400, 700, and 2600 nm; Deco Island Gold Co. Ltd., China) were mixed with the CNF dispersion at a w/v ratio of 8:10, along with sintering agents, including propane sulfone, citric acid, gluconolactone, and aspartic acid (2 wt %; Sigma-Aldrich Co. Ltd., USA), using a vortex mixer at room temperature. Then, Zn powders (2600 nm; Deco Island Gold Co. Ltd., China) were mixed

with the CNF dispersion at various w/v ratio with different concentration citric acid (1 to 4 wt %). Mo ink was prepared by mixing Mo powders (1500 nm; Deco Island Gold Co. Ltd., China) with the CNF dispersion at a w/v ratio of 6:10, also using a vortex mixer at room temperature. The CNF/ALG electrolyte ink was prepared similarly, using a 1:1 volume ratio of CNF to ALG.

Mo/PLCL inks

Mo powders (500 nm; 3A Chem Co. Ltd., China), PLCL, and dichloromethane were mixed at a w/v ratio of 6:1:8 by a magnetic mixer (IKA Co. Ltd., Germany) at room temperature to achieve Mo/PLCL inks.

Materials characterization

The rheological properties of the functional inks were evaluated at room temperature using a rheometer (Kinexus Pro+, Netzsch Co. Ltd., Germany) equipped with a 20-mm parallel plate geometry. The gap between the upper and lower plates was fixed at 1 mm for all measurements. Apparent viscosity was measured across a shear rate range of 10^{-1} to 10^3 s^{-1} . Viscoelastic behavior was assessed via oscillatory time sweep experiments, conducted at a constant frequency of 1 Hz with oscillatory strain increasing from 10^{-1} to $10^3\%$. Electrical conductivity was determined using a four-point probe measurement system (Keithley 2700, Keithley Co. Ltd., USA). The thickness of the printed films was controlled to allow accurate conversion of sheet resistance into bulk conductivity. To minimize oxidative interference in the Zn/CNF ink, conductivity measurements were further validated via linear sweep voltammetry (0 to 100 mV, 1 mV s^{-1}). The crystalline structure of printed Zn/CNF samples was characterized using XRD (D/max-2550, Rigaku Co. Ltd., Japan). Surface morphology and microstructure of the printed samples were observed using a scanning electron microscope (GEMINI SEM 500, Zeiss Co. Ltd., Germany).

3D printing and performance evaluation of biodegradable battery and bioimpedance electrodes

The CNF-based inks were used to fabricate the anode, cathode, and electrolyte via extrusion-based 3D printing using a multinozzle 3D printer (SunP Biomaker 2, SunP Biotech Co. Ltd., China). The Mo/PLCL ink was used to fabricate the biodegradable Mo/PLCL electrodes. Each ink was loaded into 5-ml disposable syringes fitted with smooth-flow tapered nozzles (inner diameter: 410 μm) secured via luer-lock connectors. 3D printing patterns were designed in Fusion 360, with the models converted into G-code to control pneumatic syringe movement. The printing was performed at a speed of 5 mm s^{-1} , with an extrusion rate of 0.7 mm s^{-1} . The center-to-center spacing between adjacent cylindrical rods was maintained at 1.0 mm. The final height of the 3D-printed battery, composed of regular cubic lattice structures, ranged from 1.0 to 1.8 mm, with successive layers deposited at a z-spacing of 0.2 mm.

The open-circuit potential and discharge behavior of the 3D-printed batteries was evaluated using a multichannel battery testing system (CT/CTE-4000, Neware Co. Ltd., China) under various conditions, including different electrolyte compositions, current densities, chemical sintering agents, and electrode thicknesses and width. Evaluation was also conducted on a 3D-printed, sandwich-structured Zn-Mo primary cell with an electrode area of 1 cm^2 and an inter-electrode spacing of 600 μm across three printed layers. To simulate in vivo conditions, the battery was tested within agarose gel matrices, with 3D-printed Mo electrodes (12 mm in length), Zn electrodes (9 mm in length), and both with width of 400 μm . The electrodes

were encapsulated with a thin layer of PLGA to enhance stability. EIS measurements were carried out for Zn/CNF- and Mo/CNF-printed electrodes using a three-electrode setup on a potentiostat (1000E, Gamry Co. Ltd., USA). The electrolyte consisted of an agarose gel containing PBS, with Ag/AgCl and platinum (Pt) serving as the reference and counter electrodes, respectively. The frequency range for EIS scans spanned from 0.1 to 10,000 Hz. EIS tests were conducted on the 3D-printed Mo/PLCL electrodes (400- μm width and 4-mm length) with a three-electrode configuration in PBS solution the same as CNF based electrodes. Impedance data were collected by sweeping the frequency from 10 Hz to 1 MHz. To assess the stability of the 3D-printed electrodes, the device was immersed in PBS (replaced daily) and maintained at 37°C in a water bath. Time-series EIS tests were performed daily using the Gamry 1000E potentiostat to monitor the electrochemical performance over time. To evaluate pH changes during operation, the Zn-Mo battery of 0.4-mm width, 10-mm length, and 0.2-mm thickness encapsulated by PLGA was immersed in PBS and discharged at a constant current density of $50 \mu\text{A cm}^{-2}$ using a multichannel battery testing system. An aliquot of 20- μl of the solution between the Zn and Mo electrodes was collected every 12 hours. The pH of each sample was immediately measured using pH test strips (range, 6.4 to 8.0, precision ± 0.3).

Fabrication of the biodegradable and multifunctional cranial device

The fabrication process involved the construction of a self-electrified scaffold for electrical stimulation and the integration of 3D-printed electrodes for bone impedance monitoring. Zn/CNF anodes and Mo/CNF cathodes (each 200 μm thick and 2 mm in diameter) were 3D-printed onto an MC/PCL substrate (200 μm thick and 5 mm in diameter), followed by encapsulation with an additional MC/PCL layer. The MC/PCL scaffold was prepared according to previously established protocols (80). Briefly, HA-mineralized collagen powder and PCL (molecular weight 300,000) were dispersed in 1,4-dioxane at a w/v ratio of 1:1:10 (grams per milliliter). After 4 hours of magnetic stirring to ensure uniform dispersion, the mixture was cast into cylindrical molds, frozen at -20°C overnight, and lyophilized to form a porous MC/PCL substrate via solvent removal. The 3D-printed bioimpedance electrodes were fabricated by printing Mo/PLCL (8-mm length and 200- μm thickness) directly onto PLCL films. PLCL films (200- μm thickness) were prepared by dissolving PLCL (Jinan Daigang Biomaterial Co. Ltd., China) in dichloromethane (CH_2Cl_2 ; Beijing Tongguang Chemical Co. Ltd., China) at a 1:10 w/v ratio. The solution was drop-cast and cured at 4°C for 12 hours to minimize bubble formation. Solvent bonding was used to integrate the Mo/PLCL electrodes on a PLCL substrate with the self-powered MC/PCL scaffold (adhesion is achieved between the PCL and MC/PCL layers), and the Mo/PLCL electrodes are attached to the FPC connector by thermo-compression using silver paste zebra paper.

Assessment of ROS generation

ROS production at the Mo electrode interface following battery discharge was evaluated using a DCFH (2',7'-dichlorodihydrofluorescein) probe derived from a commercial DCFH-DA kit (Solarbio Science & Technology Co. Ltd., China). The probe was first hydrolyzed under alkaline conditions by dissolving it in an aqueous NaOH solution (pH 10) and incubating at room temperature for 20 min. The pH of the resulting solution was then adjusted to physiological levels using PBS, yielding the working probe solution for subsequent

analysis. OCV testing of the cell was conducted in a PBS environment. At predetermined time points (10, 30, 60, and 120 min) during the OCV test, the solution adjacent to the Mo electrode were sampled and immediately mixed with the prepared DCFH solution. Fluorescence emission spectra were recorded using a fluorometer (excitation/emission: 488/525 nm) to quantify ROS levels.

Finite element analysis

The 3D FEA model of the printed Zn-Mo battery was performed. The model geometry and dimensions were consistent with the experimental configuration. In the simulation, Zn was served as the anode and Mo as the cathode. The potentials of the cathode (outer ring electrode) and anode (inner ring electrode) were set to 0.4 and -0.4 V based on experimental measurements, respectively. The surrounding electrolyte was assumed a dielectric medium mimicking cranial tissues, with a relative permittivity of 80. The boundary potential was fixed at 0 V.

Influence of biodegradable batteries on hBMSCs

To assess the effects of the battery, four experimental groups were established: hBMSCs cultured on regular cell culture dish (control group), hBMSCs cocultured with the Zn-Mo battery device (battery group), with the Zn electrode alone (Zn group), or with the Mo electrode alone (Mo group). The electrodes were encapsulated with a thin layer of PLCL to control ion release. Each foil electrode has dimensions of 20 μm in thickness and 200 μm by 15 mm in width and length, respectively. hBMSCs was cultured in minimum essential medium (MEM) complete medium, consisting of 89% MEM, 10% fetal bovine serum, and 1% penicillin-streptomycin. The biocompatibility of the battery was assessed using a Calcein-AM/PI live/dead staining kit (Beyotime Biotechnology, China). Viable cells were labeled with Calcein-AM (green fluorescence), while dead cells were stained with PI (red fluorescence). Fluorescent images were captured using an inverted fluorescence microscope (Olympus IX73, Japan). To evaluate the osteogenic potential of the battery, hBMSCs were seeded into 6- or 12-well culture plates. Upon reaching 70 to 80% confluency or successful adhesion, materials from each experimental group were introduced and fixed to the well walls, followed by the addition of osteogenic induction medium. For ALP and ARS staining, hBMSCs were treated with different material groups for 7 or 14 days in osteogenic medium. The experiments comprised the following groups: control group (osteogenic medium only), Zn group (exposed to a Zn electrode or its leachate, respectively), Mo group (exposed to a Mo electrode or its leachate, respectively), Mix group (exposed to a combined leachate of Zn and Mo electrodes), and battery group (directly cocultured with the Zn-Mo battery). ALP activity and calcium deposition were visualized using an ALP staining kit and ARS (Solarbio Science & Technology Co. Ltd., China), respectively, according to the manufacturer's protocols. To further validate the pro-osteogenic roles of elevated ROS and Ca^{2+} levels induced by the Zn-Mo battery, four experimental groups were established with four replicates each: hBMSCs cultured on regular cell culture dish (control group), hBMSCs cocultured with Zn-Mo battery (battery group), hBMSCs cocultured with the battery and 50 μM NIF (battery + NIF group), and hBMSCs cocultured with the battery and 500 μM NAC (battery + NAC group). When hBMSCs cultured in six-well plates reached 70 to 80% confluence, the prepared NIF or NAC working solutions were added to the culture medium of the corresponding groups. ALP and ARS staining were performed on

days 7 and 14, respectively, following the protocols described above. Quantitative analysis of staining intensity was performed using ImageJ, with average integrated density used to evaluate ALP expression and mineralization levels.

To assess intracellular calcium ion (Ca^{2+}) dynamics, the Fluo-4 AM probe (Beyotime Biotechnology, China) was used. The working solution was prepared at a concentration of 2 μM . hBMSCs were rinsed with PBS three times and incubated with Fluo-4 AM at 37°C for 30 min, followed by three additional PBS washes. The cells were then incubated for an additional 30 min to allow complete de-esterification of the dye. Following material treatment, Ca^{2+} fluorescence was recorded using an inverted fluorescence microscope (Olympus IX73, Japan). Fluorescence intensity changes were quantified as $\Delta F/F_0 = (F - F_0) / F_0$, where F_0 represents baseline fluorescence at time zero.

For phalloidin staining, cells were rinsed with PBS and fixed in 4% formaldehyde for 15 min. After three PBS washes, they were incubated with fluorescein isothiocyanate-conjugated anti-adhesion plaque protein antibody for 1 hour, followed by staining with rhodamine-labeled phalloidin for 45 min and DAPI counterstaining for 5 min. Fluorescence imaging was performed using a fluorescence microscope (Olympus IX73, Japan).

For Western blotting, hBMSCs were lysed on ice for 30 min using radioimmunoprecipitation assay buffer (Thermo Fisher Scientific, USA) supplemented with phosphatase and protease inhibitors. Lysates were centrifuged at 12,000g for 30 min at 4°C, and supernatants were collected for total protein extraction. Protein concentrations were determined using the Pierce BCA Protein Assay Kit (Thermo Fisher Scientific, USA). Equal amounts of protein (10 μg) were mixed with 5 $\times$ SDS loading buffer, denatured at 100°C for 5 min, and separated by 10% SDS-polyacrylamide gel electrophoresis, followed by transfer onto nitrocellulose membranes (Bio-Rad, USA). Membranes were blocked with 5% bovine serum albumin in TBS-T (tris-buffered saline with 0.1% Tween-20) for 2 hours at room temperature and incubated overnight at 4°C with primary antibodies against Runx2, Wnt5a/b, Bmp2, β -tubulin, and glyceraldehyde-3-phosphate dehydrogenase (Abcam or Invitrogen), diluted per the manufacturer's instructions. After three washes with TBS-T, the membranes were incubated with horseradish peroxidase-conjugated secondary antibodies (Santa Cruz Biotechnology, 1:3000) for 2 hours at room temperature. Protein bands were visualized using SuperSignal West Pico PLUS substrate (Thermo Fisher Scientific) and imaged with a Tanon 5200 system (Tanon Life Sciences, China).

In vivo evaluation of cranial bone defect repair

Male SD rats (8 weeks old, ~300 g) were obtained from VitalRiver Laboratories (Beijing, China). All of the experimental procedures involving animals were performed in accordance with Guides for the Care and Use of Laboratory Animals from the Chinese Ministry of Public Health and the US National Institutes of Health (approval document number: 18-SX1, Laboratory Animal Research Center, Tsinghua University). All designs and implements of animal experiment followed the principle of "3Rs" (reduction, replacement, and refinement). Before surgery, the animals were anesthetized via intraperitoneal injection of sodium pentobarbital (3%, 30 mg/kg body weight). A midline scalp incision was made to expose the cranial bone by blunt dissection of the overlying soft tissue. Bilateral full-thickness cranial defects (5-mm diameter) were created using a precision dental drill. Rats were randomly assigned into four groups

($n = 4$ per group). A mineralized MC/PCL scaffold (400 μm thick and 5-mm diameter) was implanted into the left-side defect as a control. The right-side defect received either the scaffold integrated with the Zn-Mo battery or a single-electrode component (Zn or Mo). Longitudinal assessment of bone regeneration was conducted using an *in vivo* micro-CT imaging system (Quantum GX2, PerkinElmer, USA) at weeks 0, 2, 4, 8, and 12 postimplantation. Micro-CT data were reconstructed and quantitatively analyzed using Quantum, CTAn, and Data Viewer software. Animals were euthanized at 4 and 12 weeks postsurgery. Cranial specimens encompassing the defect sites were harvested, fixed in paraformaldehyde, decalcified in EDTA (Thermo Fisher Scientific), and subjected to graded ethanol dehydration. Histological analyses, including H&E, Masson's trichrome, MB, and Sirius Red staining, were performed following standard protocols. Immunohistochemical staining for bone-related markers, including Bmp2, osteocalcin, bone sialoprotein (BSP), and collagen type I alpha 1 (Col1A1) was conducted to assess osteogenic activity. For RNA sequencing, at 3 weeks postimplantation, cranial tissue samples were collected and stored at -80°C until RNA sequencing (RNA-seq). RNA-seq samples were extracted and sequenced on Novogene Co. Ltd. Quality control was performed by FastQC and read alignment as well as gene expression quantification of raw sequencing data were processed using STAR. All data analysis was performed using R (version 4.2.3). The DaMiRseq package was used to normalize the original levels of gene expressions and identify surrogate variables to eliminate the influence of batch differences between samples. Identification of DEGs between the control and battery groups was conducted with the DESeq2 package, and they were defined with a threshold of a P value <0.05 and a fold change >1.5 to show heatmaps. Functional enrichment analysis (GO and KEGG) was performed using the clusterProfiler package. This analysis focused on up-regulated DEGs, which were selected using a threshold of P value <0.05 and fold change >1.25 to indicate biological pathways and processes exhibiting a general up-regulatory trend. In addition, GSEA was implemented through the clusterProfiler package, with pathways considered significantly enriched at a threshold of a q value <0.05 and an |NES (Normalized Enrichment Score)| ≥ 1.5 .

Simulate impedance change of early bone regeneration with *in vitro* models

To simulate the biochemical environment and assess the associated impedance changes during the early stages of bone healing, ALG-based hydrogels incorporating varying concentrations of collagen and nano-HA (nHA) were fabricated. Sodium ALG (medium viscosity; Sigma-Aldrich, USA) was dissolved in deionized water at a concentration of 2% (w/v). This solution was then mixed with either type I collagen (rat tail-derived; Sigma-Aldrich, USA) at concentrations of 10, 15, 20, or 25 wt %, or with nHA (50 nm, Shanghai Macklin Biochemical Technology Co., China) at concentrations of 10, 30, or 50 wt %. Cross-linking was achieved by immersing the mixtures in a 2-wt % CaCl_2 solution, resulting in hydrogel discs (5-mm diameter and 1-mm thickness). Electrical impedance measurements were conducted on the hydrogels using 3D-printed electrodes and EIS.

In vivo impedance monitoring of bone healing

Male SD rats (8 weeks old, ~ 300 g) were anesthetized via intraperitoneal injection of 3% sodium pentobarbital (30 mg/kg body weight). Eight-week-old male SD rats were selected on the basis of established protocols in cranial defect models to minimize biological

variability and ensure reproducibility (13, 15). Critical-sized full-thickness cranial defects (5-mm diameter) were created using a cylindrical dental drill. The cranial device was inserted into the cranial defects (5 mm in diameter and 1 mm in thickness) on top of the dura mater, and bone nails were used to secure the extended wings for device immobilization. Suturing the outer periosteum and scalp further stabilizes the device. These procedures help prevent discomfort and recording instability caused by device movement during rat activity. In addition, the rats were disinfected and cleaned daily for 1 week after surgery to prevent infection. The multifunctional cranial device was implanted into the defect site in the experimental group, while 3D-printed electrodes alone without the self-electrified scaffold were implanted in the control group ($n = 3$ per group). The FPC connector was routed subcutaneously along the spine to the dorsal surface for continuous monitoring. The progress of bone regeneration at weeks 1, 2, 3, and 4 was monitored via EIS using a Gamry Interface 1000E potentiostat. Impedance data were collected by sweeping frequencies from 100 Hz to 1 MHz, using a 5-mV sinusoidal excitation signal at 0-V bias relative to the open-circuit potential. Animals were euthanized at 4 weeks postsurgery. Cranial tissues encompassing the defect regions were harvested, fixed in paraformaldehyde, decalcified in EDTA (Thermo Fisher Scientific), and dehydrated through graded ethanol series. Standard histological analyses, including H&E, Masson's trichrome, and Sirius Red staining, were performed to evaluate the corresponding healing status.

Equivalent circuit model of EIS data for bone healing assessment

The equivalent circuit models were adapted from previously established frameworks (67). Within this analog representation, the interfacial impedance of the electrical double layer was modeled using a constant phase element (CPE_{dl}), serially connected to components representing tissue electrophysiology. To account for parasitic signal coupling through insulated conductors, a parallel capacitance was incorporated into the circuit. The biological milieu was modeled as a parallel resistor-capacitor (R_t and C_t), followed by a resistor R_{2t} in series. R_t corresponds to intracellular ionic conduction, R_{2t} denotes extracellular ionic conduction, and C_t represents the double layer cell membranes. Temporal impedance data were analyzed using a custom MATLAB algorithm that iteratively fit circuit parameters to the experimental data via nonlinear optimization.

Evaluation of *in vitro* and *in vivo* biodegradation

For *in vitro* degradation studies, the self-electrified scaffold and Mo/PLCL electrodes were immersed in PBS (pH 7.4) and incubated at 60°C in a water bath. The PBS solution was refreshed daily, and photographs were taken at different time points to monitor degradation progression.

For *in vivo* biodegradation and biocompatibility evaluation, the self-electrified scaffold and Mo/PLCL electrodes were implanted subcutaneously in SD rats. Following anesthesia, a small incision was made on the dorsal skin, and the devices were inserted into the subcutaneous space before suturing. Photographs were taken every 4 weeks to monitor degradation *in vivo*. Twelve weeks postimplantation, major organs (heart, liver, spleen, lungs, kidneys, and brain) were harvested for histological examination by H&E staining to assess biocompatibility.

Statistical analysis

Statistical analyses were performed using GraphPad Prism (version 10.0). Two-group comparisons were conducted using unpaired t

tests, while multiple-group comparisons were analyzed by one-way analysis of variance (ANOVA) followed by Tukey's post hoc test or Dunnett's T3 test, as appropriate. Statistical significance was defined as $*P < 0.05$, $**P < 0.01$, $***P < 0.001$, and $****P < 0.0001$. Pairwise comparisons not displayed in the graphs were not statistically significant ($P > 0.05$). Data are reported as means $\pm$ standard deviation.

Supplementary Materials

This PDF file includes:

Figs. S1 to S29

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3D-Printed biodegradable and multifunctional cranial device for bone repair and healing assessment

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Supplementary Materials for
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and healing assessment**

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This PDF file includes:

Figs. S1 to S29

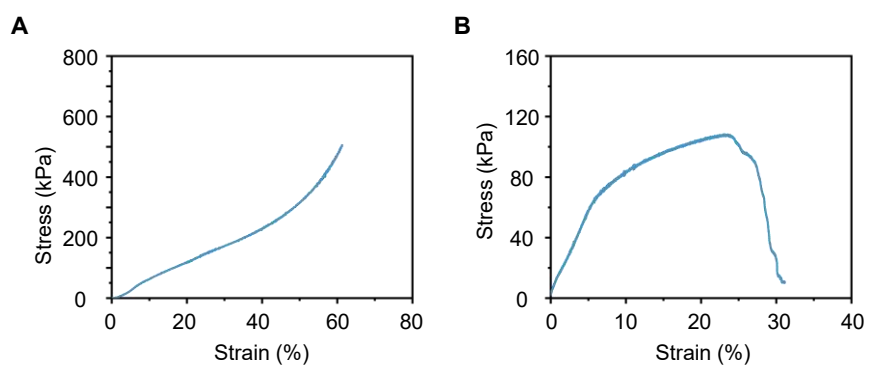


Figure S1. (A) tensile stress–strain curve and (B) compression stress–strain curve of the MC/PCL scaffold.

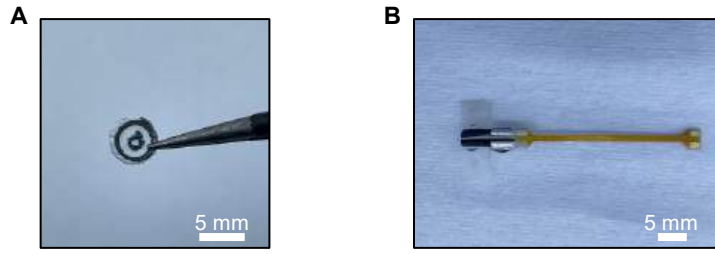


Figure S2. Photos of the biodegradable components of the cranial devices. (A) Photograph of 3D printed self-electrified scaffold, with Zn-Mo battery on the MC/PCL substrate. (B) Photograph of 3D printed Mo/PLCL bioimpedance electrodes.

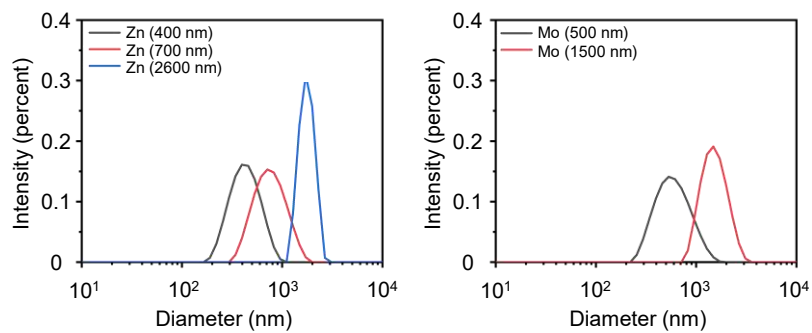


Figure S3. Particle size distribution of Zn and Mo particles.

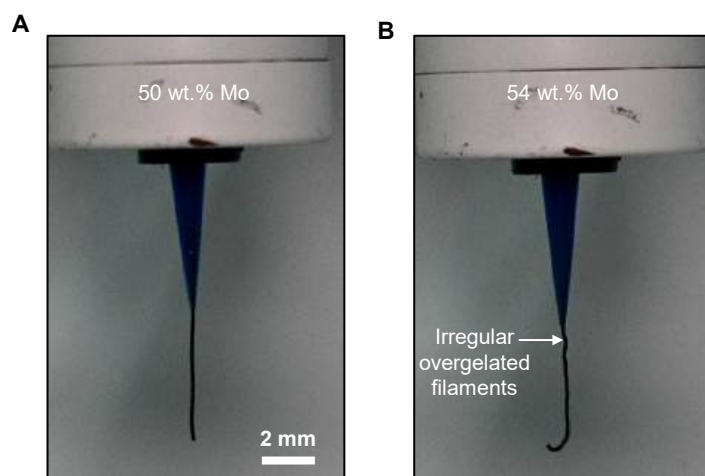


Figure S4. Photos of the Mo/CNF ink during the printing process. (A) Smooth filaments with 50 wt.% Mo; **(B)** Irregular overgelated filaments with 54 wt.% Mo.

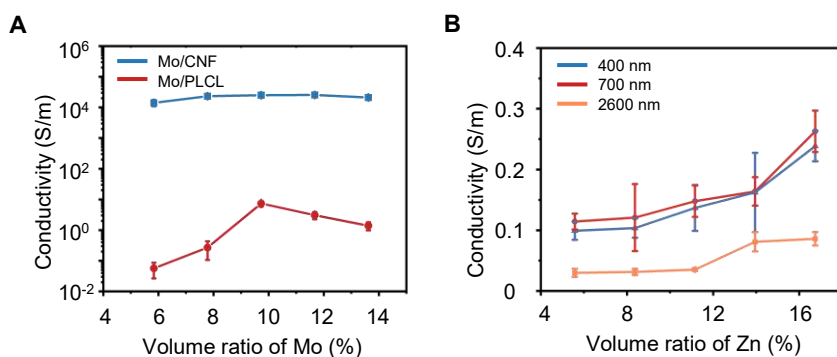


Figure S5. The relationship between electrode conductivity and the volume fraction of metal particles. (A) Conductivity of Mo/CNF and Mo/PLCL inks as a function of volume ratio of Mo. (B) Conductivity of Zn ink as a function of volume ratio of Zn with 2 wt.% citric acid sintering agents.

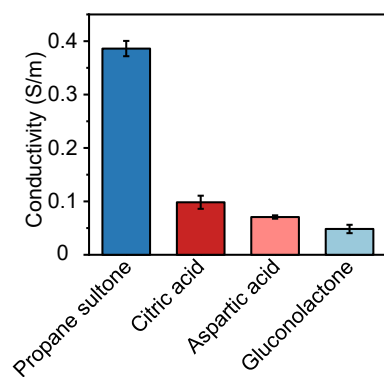


Figure S6. Conductivity of Zn ink with different sintering agents.

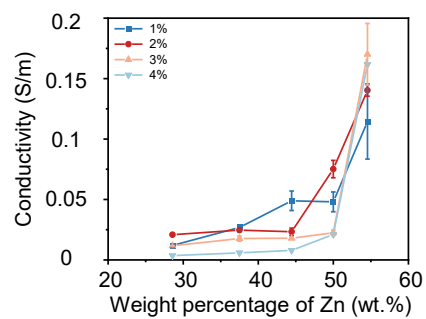


Figure S7. Electrical conductivity of inks with different Zn (2600 nm) contents treated with 1 wt.%, 2 wt.%, 3 wt.% and 4 wt.% citric acid.

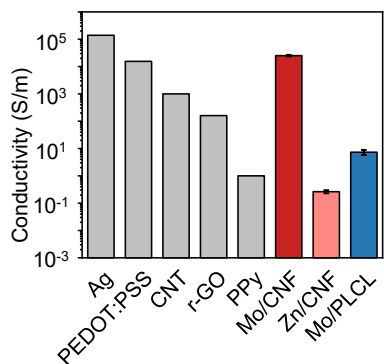


Figure S8. Comparison of the conductivity of reported 3D printing inks with current work, including Ag/hydrogel microparticle ink (18), pure PEDOT:PSS ink (33), proteoglycan-like gel/CNT ink (36), alginate-gelatin/r-GO ink (35) and alginate-gelatin/polypyrrole:PSS ink (34).

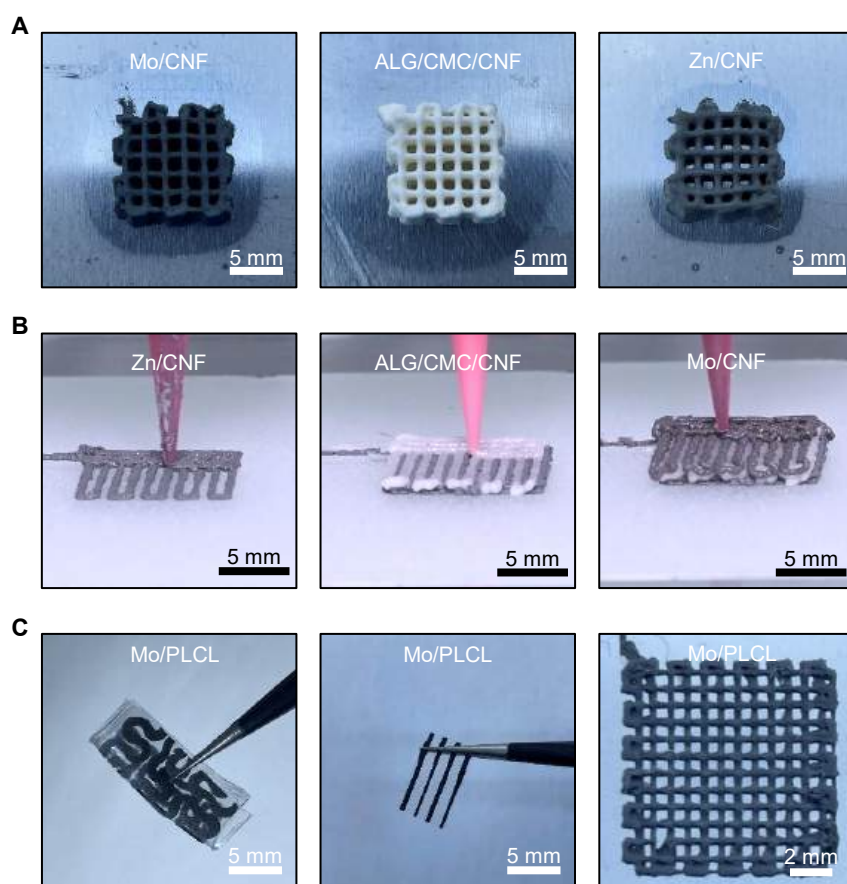


Figure S9. Photographs of 3D printed samples. (A) Photographs of 3D printed Zn, Mo, ALG/CNF. **(B)** 3D printing process of fully biodegradable primary Zn-Mo battery. **(C)** Photographs of 3D printed Mo/PLCL electrodes.

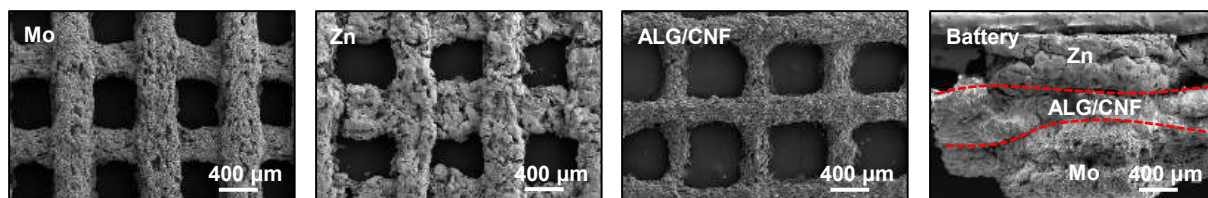


Figure S10. SEM images of the fully biodegradable 3D printed Zn–Mo battery.

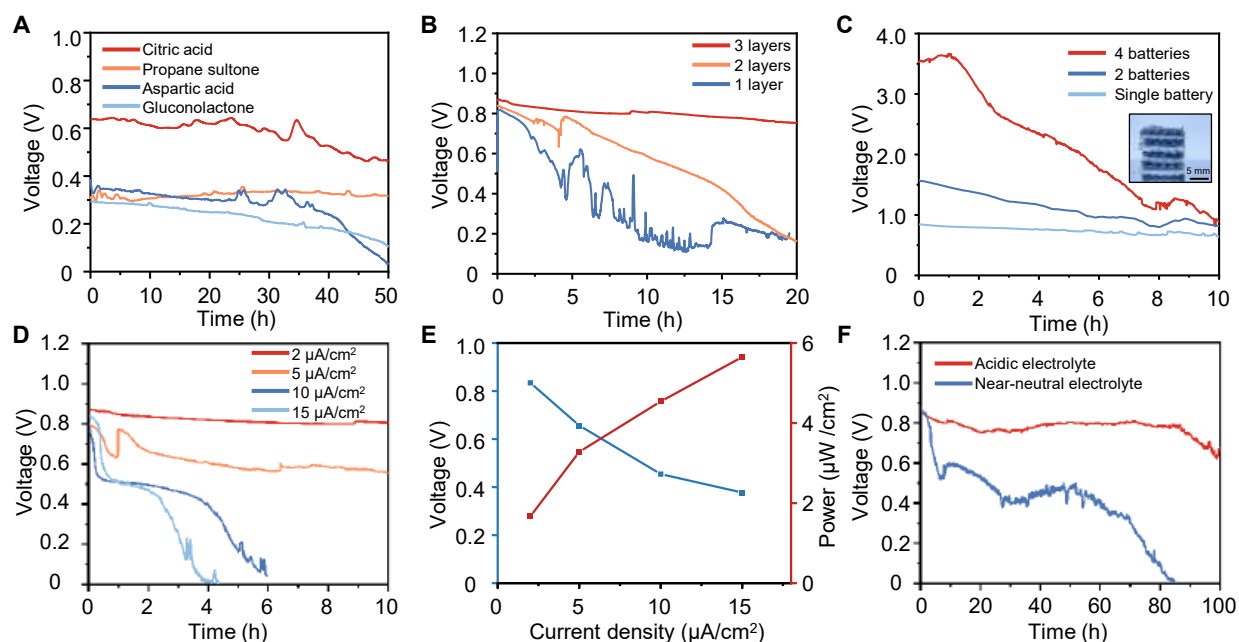


Figure S11. Discharge characteristics of the fully biodegradable 3D printed Zn-Mo battery. (A) Discharge behavior of 3D printed Zn-Mo batteries, with Zn electrodes sintered by citric acid, propane sultone, aspartic acid, and gluconolactone. Discharge current density: $2 \mu\text{A cm}^{-2}$. (B) Discharge behavior of batteries with varying electrode thickness. Discharge current density: $2 \mu\text{A cm}^{-2}$. (C) Discharge behavior of batteries connected in series. Discharge current density: $2 \mu\text{A cm}^{-2}$. (D) Discharge characteristics of batteries with varying discharge current densities. (E) Battery voltage and power as a function of discharge current densities. (F) Discharge behavior of 3D printed Zn-Mo batteries with acidic electrolyte (pH 6.5) and near-neutral electrolyte (pH 7.3). Discharge current density: $2 \mu\text{A cm}^{-2}$.

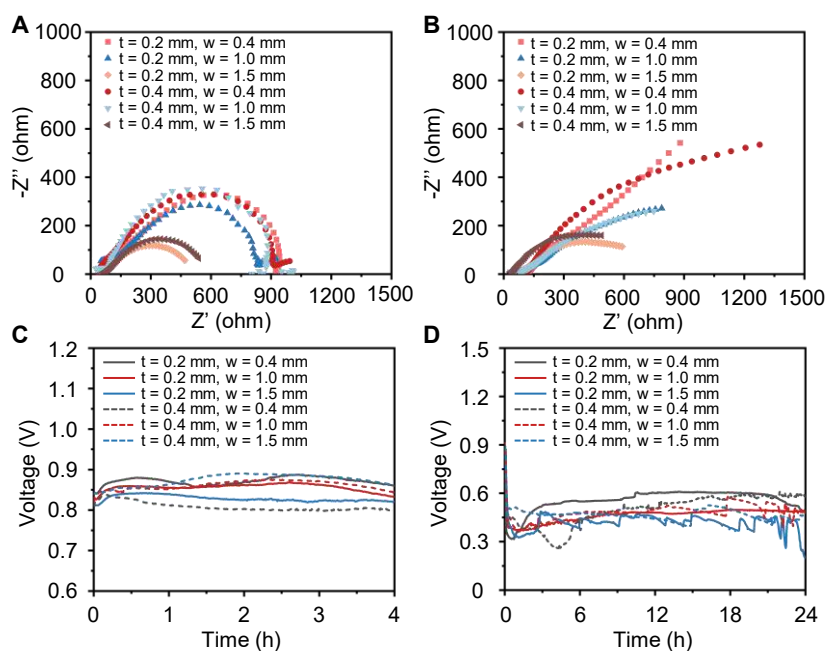


Figure S12. Electrochemical characterization of the 3D printed Zn-Mo battery. (A) EIS of the 3D-printed Zn electrode with different electrode dimensions in agarose gel (thickness: 0.2 mm, 0.4 mm; width: 0.4 mm, 1.0 mm, 1.5 mm; length: 10 mm). (B) EIS of the 3D-printed Mo electrode with different electrode dimensions in agarose gel. (C) The open-circuit potential of batteries with different electrode dimensions in agarose gel. (D) Galvanostatic discharge curve of the batteries with different electrode dimensions at a current density of $50 \mu\text{A cm}^{-2}$ in agarose gel.

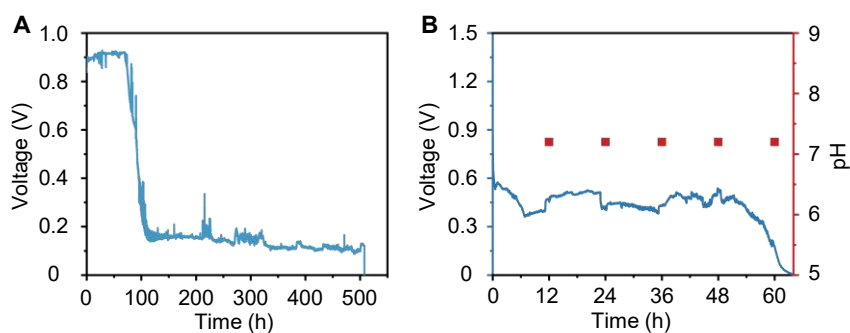


Figure S13. Open-circuit potential and pH monitoring of the 3D printed Zn-Mo battery. (A) The open-circuit potential of the Zn-Mo battery in agarose gel simulating physiological conditions with Mo electrodes 12 mm in length, Zn electrodes 9 mm in length, and both electrodes 400 μm in width and 200 μm in thickness. (B) Local pH values near the Zn electrode during discharge at a current density of $50 \mu\text{A cm}^{-2}$ in PBS with both electrodes 0.4 mm in width, 0.2 mm in thickness and 10 mm in length. The pH of the collected solution is measured using pH test strips with a sensitivity of 0.3.

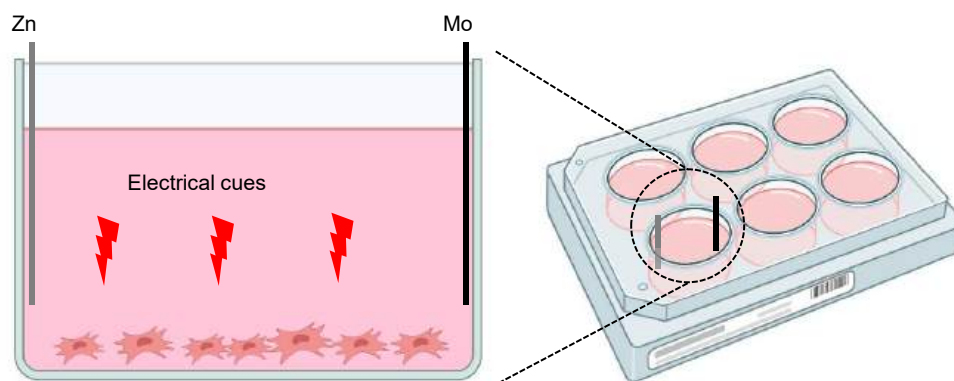


Figure S14. Schematic illustration of the in vitro experimental setup for evaluating the effects of Zn-Mo batteries on hBMSCs. Created in BioRender. xibo, X. (2026) <https://BioRender.com/yiduxtl>.

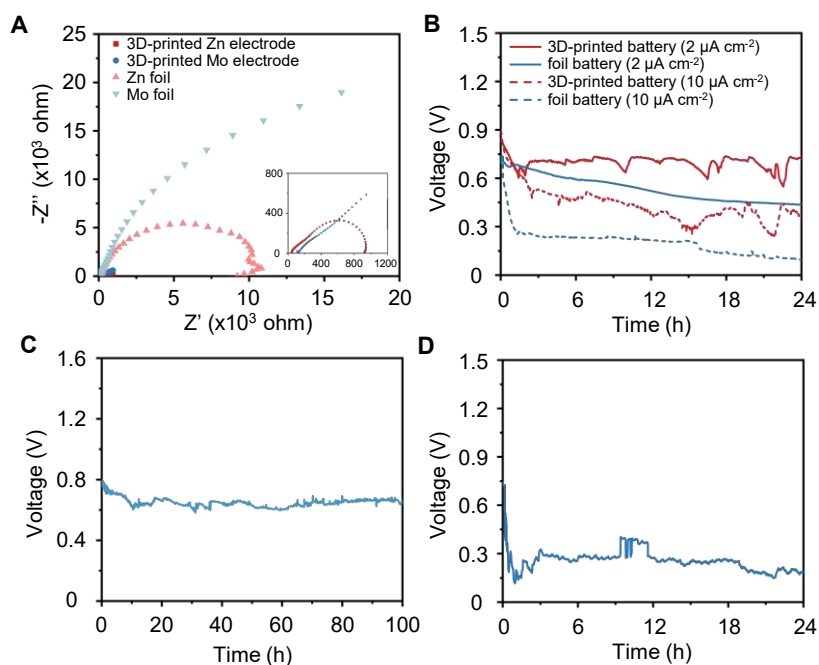


Figure S15. The Electrochemical characterization of 3D-printed Zn-Mo battery and foil Zn-Mo battery. (A) EIS of 3D-printed electrodes and foil electrodes with 0.4 mm in width and 10 mm in length. (B) Galvanostatic discharge curves of the 3D-printed Zn-Mo battery and the foil Zn-Mo battery at a current density of 2 and $10 \mu\text{A cm}^{-2}$ in agarose gel simulating physiological conditions. (C) Open-circuit voltage of the foil Zn-Mo battery in cell culture medium. (D) Galvanostatic discharge curves of the Zn-Mo battery with foil electrodes at a current density of $10 \mu\text{A cm}^{-2}$ in PBS.

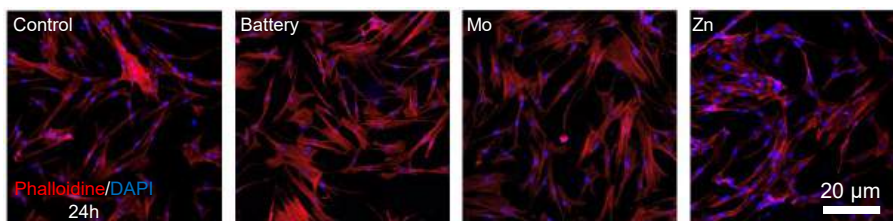


Figure S16. Phalloidine/DAPI fluorescence images indicating cell morphology. Immunohistochemical staining: actin cytoskeleton (Phalloidine, red) and nuclei (DAPI) blue). $n = 3$ independent experiments.

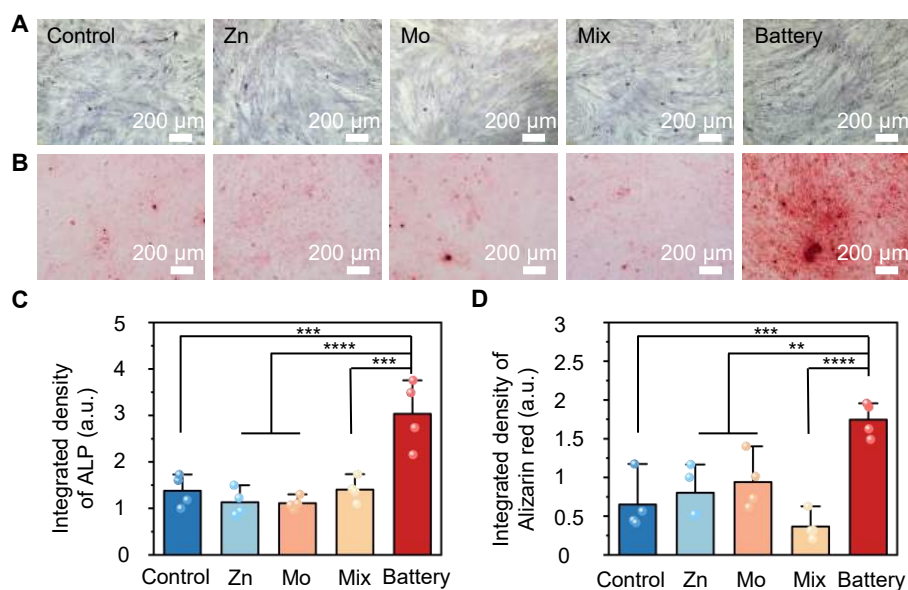


Figure S17. The effect of metal ions on osteogenesis. (A) ALP staining of hBMSCs on day 7. (B) Alizarin red staining of hBMSCs on day 14. (C) Quantitative evaluation of the ALP staining integrated density on day 7. (D) Quantitative evaluation of the Alizarin red staining integrated density on day 14. In (C) and (D) data are means $\pm$ SD. One-way ANOVA with Tukey's multiple comparisons test were used for the statistical analysis. **P < 0.01, ***P < 0.001, ****P < 0.0001. In (A) and (B) n = 4 independent experiments.

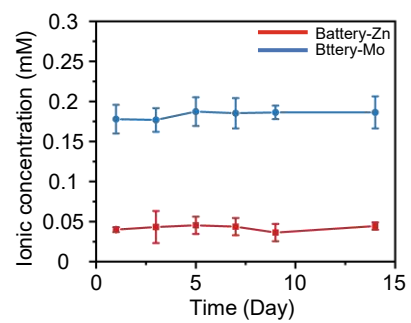


Figure S18. ICP-OES analysis results of Mo and Zn release in cell culture medium.

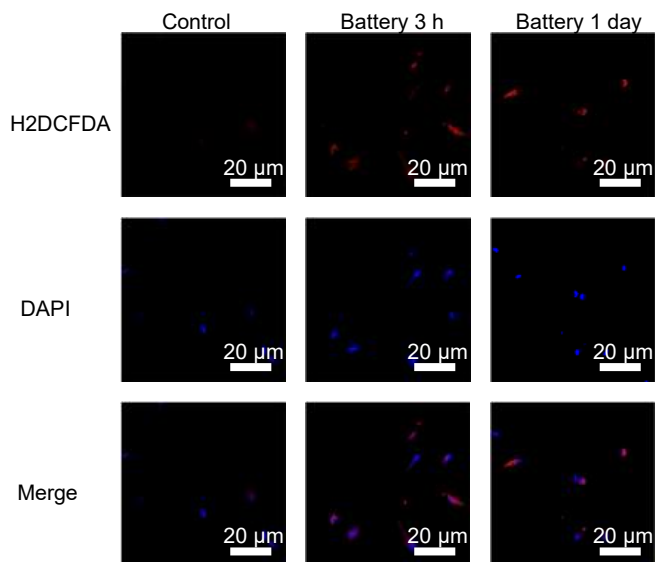


Figure S19. Intracellular reactive oxygen species levels of hBMSCs cultured with Zn-Mo batteries for 3 hours and 1 day. Immunohistochemical staining: ROS (H2DCFDA) red) and nuclei (DAPI) blue). $n = 3$ independent experiments.

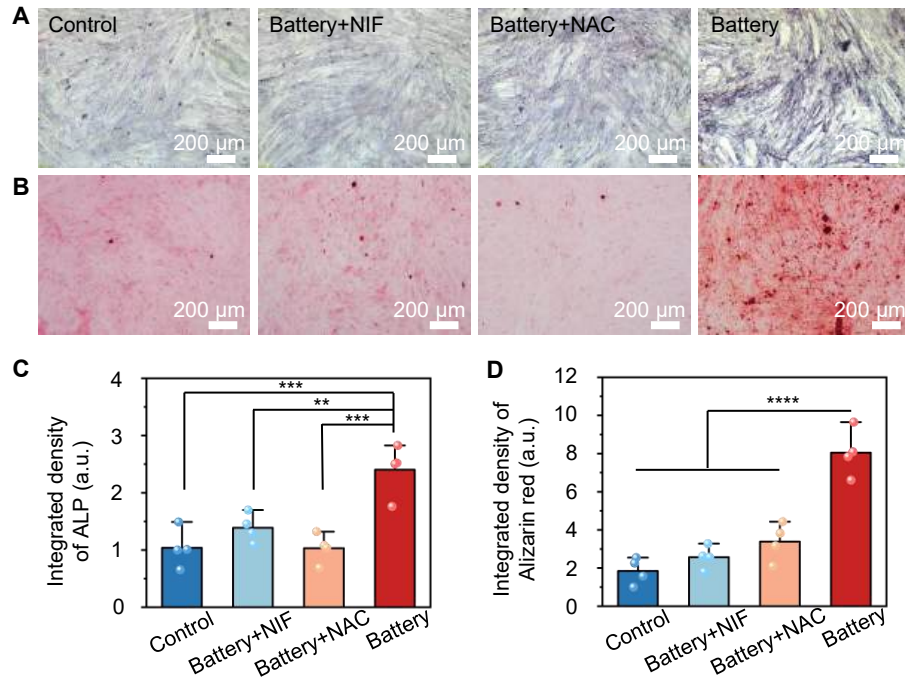


Figure S20. The effect of ROS and Ca^{2+} levels on osteogenesis with ROS scavengers or calcium channel blockers. (A) ALP staining of hBMSCs on day 7. (B) Alizarin red staining of hBMSCs on day 14. (C) Quantitative evaluation of the ALP staining integrated density on day 7. (D) Quantitative evaluation of the Alizarin red staining integrated density on day 14. In (C) and (D) data are means $\pm$ SD. One-way ANOVA with Tukey's multiple comparisons test were used for the statistical analysis. ** $P < 0.01$, * $P < 0.001$, **** $P < 0.0001$. In (A) and (B) $n = 4$ independent experiments.**

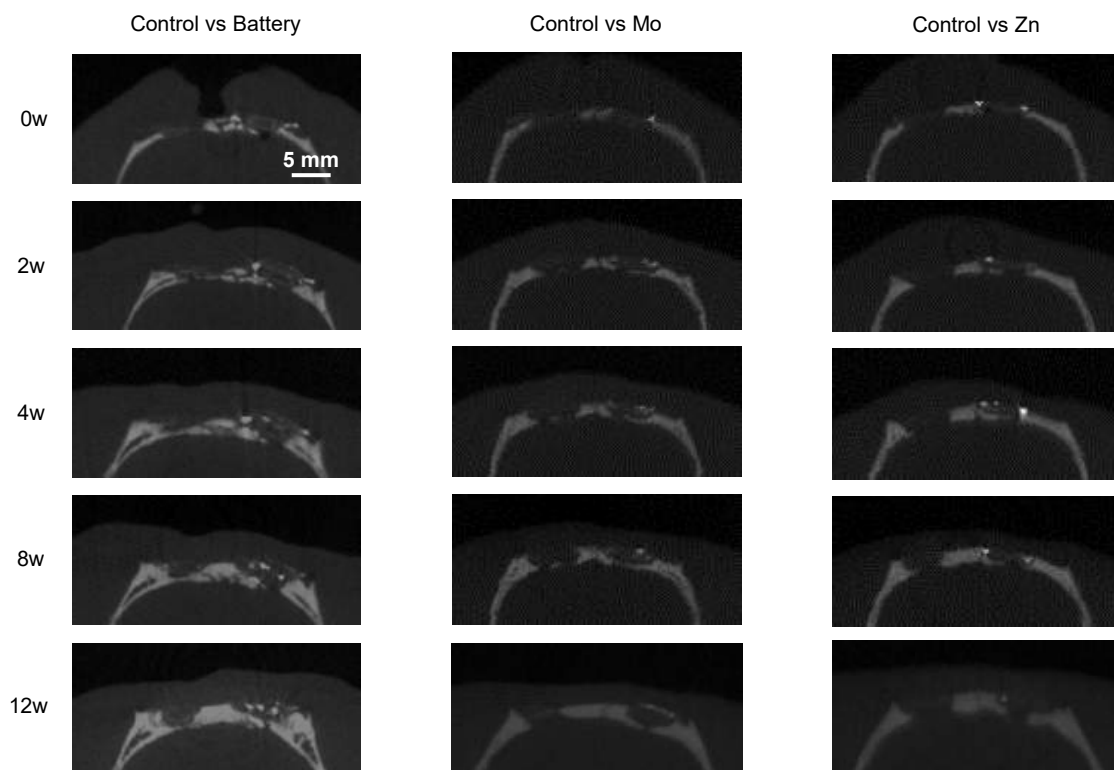


Figure S21. Radiographic assessment of cranial defect repair from 0 to 12 weeks postimplantation using micro-CT imaging (The 12w images are also presented in Fig. 4D). $n = 4$ Independent experiments.

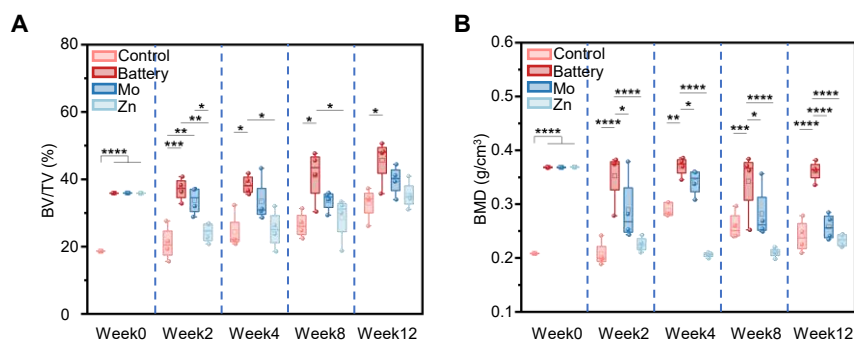


Figure S22. Statistical results of BV/TV (A) and BMD (B) for newly formed bone tissue. The box plot presents the median (center line), lower quartile (lower border), upper quartile (upper border), maximum (upper whisker) and minimum (lower whisker), which are ≤ 1.5 times the interquartile range. One-way ANOVA with Tukey's multiple comparisons test were used for the statistical analysis. * $P < 0.05$, ** $P < 0.01$ and *** $P < 0.001$. $n=4$ independent experiments.

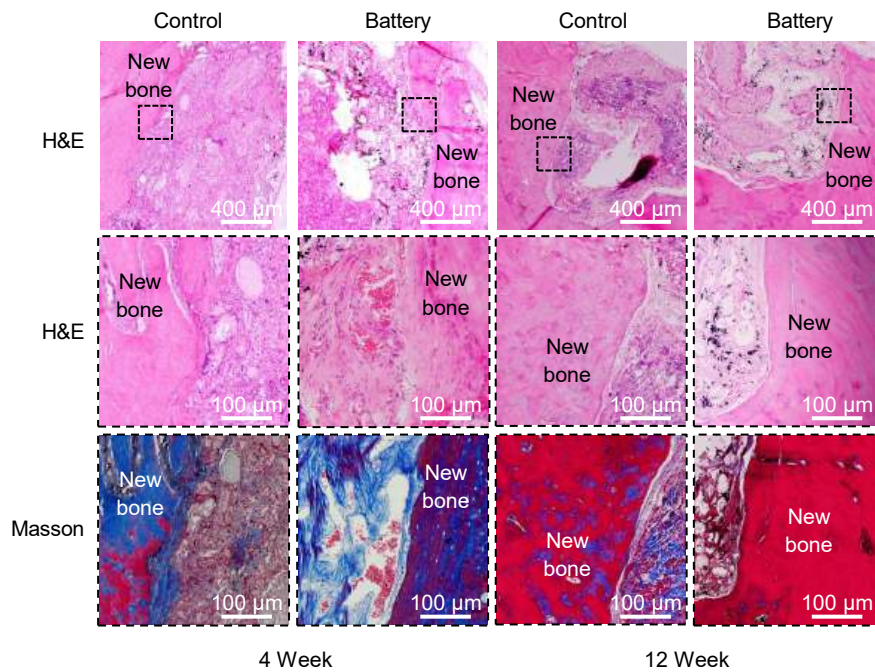


Figure S23. Transverse sections of regenerated bone tissues stained with H&E and Masson's trichrome for the control and battery groups at week 4 and 12. $n = 4$ Independent experiments.

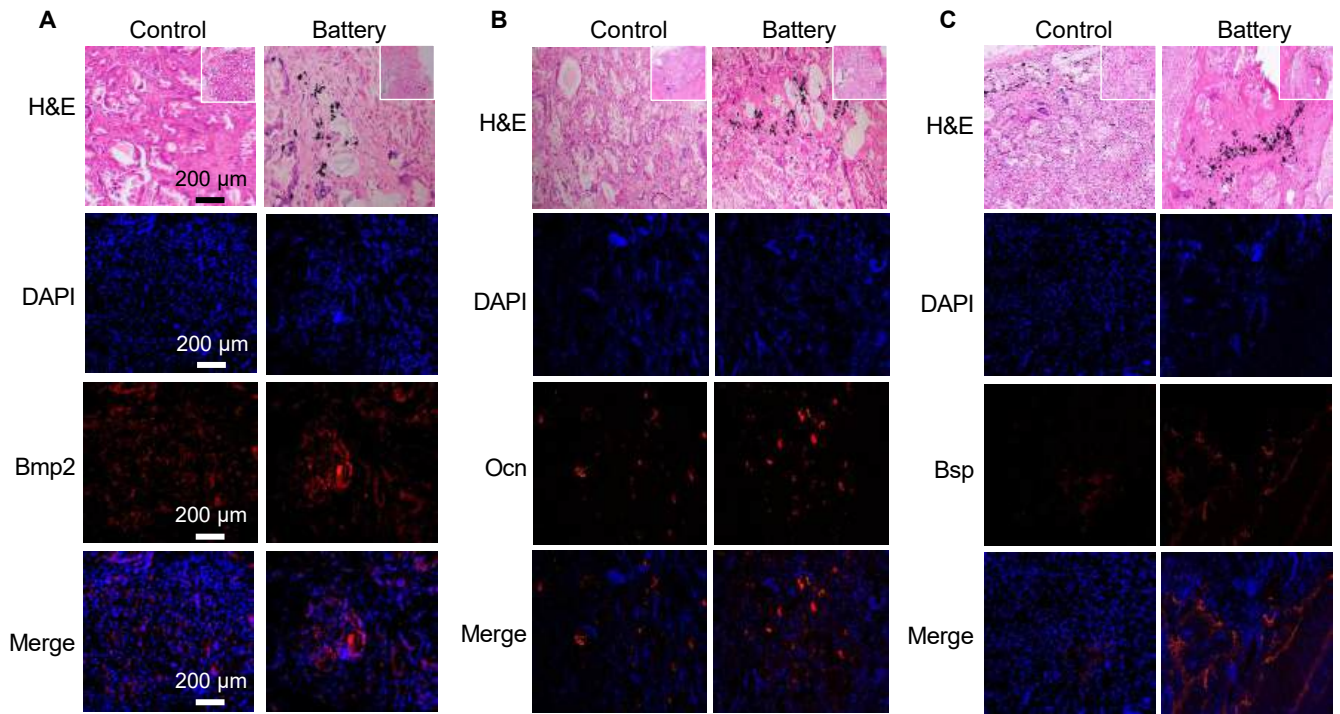


Figure S24. Histological staining images of regenerated bone tissues with self-electrified scaffolds 4 weeks postimplantation. (A) Bmp2. (B) Ocn. (C) Bsp. Immunohistochemical staining: Bmp2 (red) in (A) Ocn (red) in (B) Bsp (red) in (C) and nuclei (DAP(I) blue). $n = 3$ independent experiments.

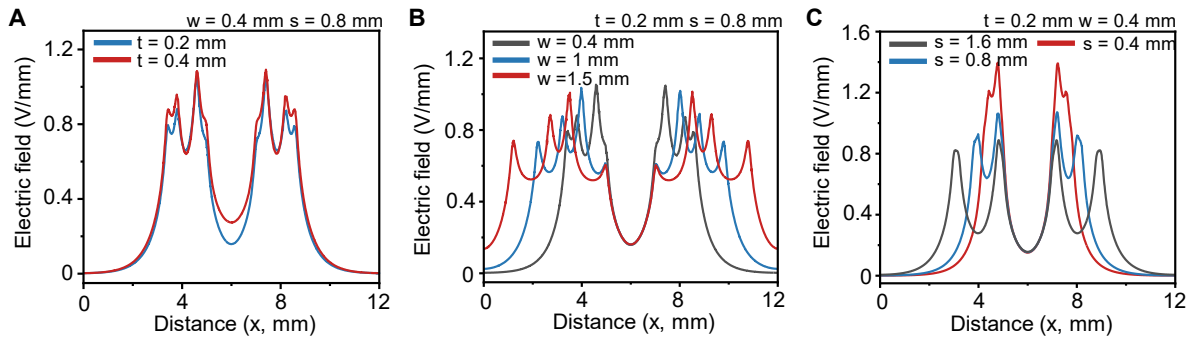


Figure S25. Finite element analysis of the electric field distribution under varying electrode geometric parameters. Simulated electric field profiles along the x-axis at 0.1 mm above the electrode surface, demonstrating the effects of **(A)** electrode thickness (t), **(B)** electrode width (w), and **(C)** inter-electrode spacing (s).

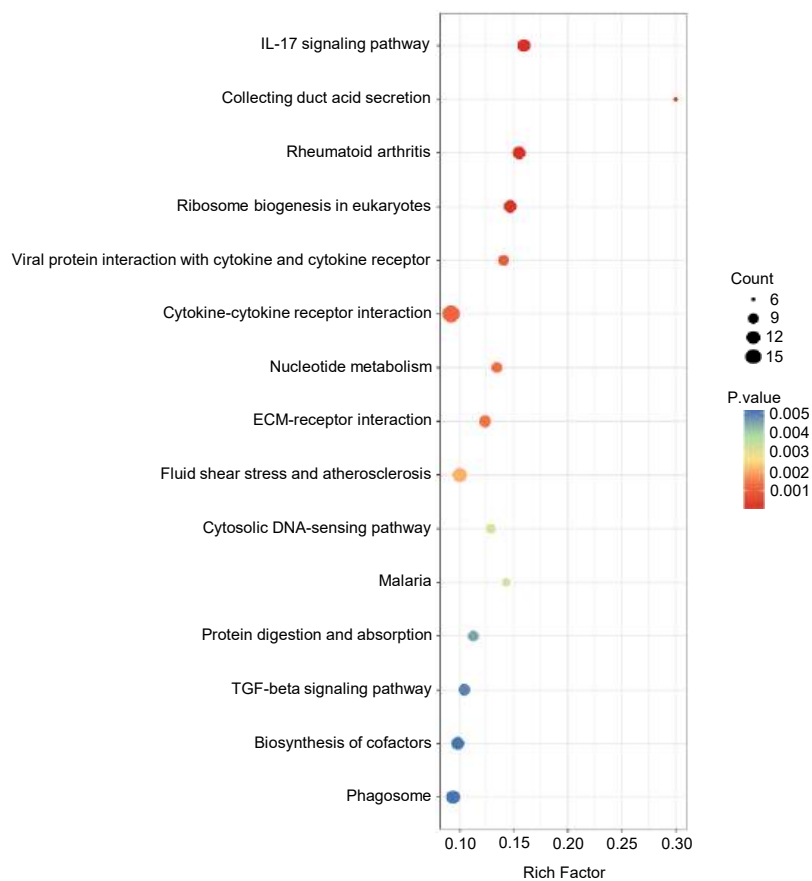


Figure S26. KEGG pathway enrichment analysis of regenerated tissues. $n = 3$ independent experiments.

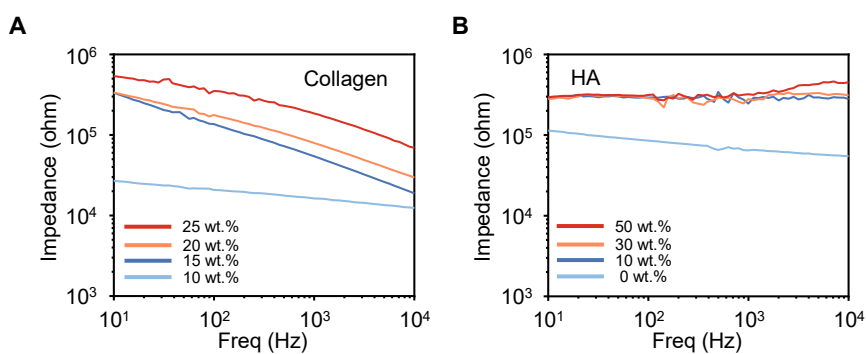


Figure S27. Changes of impedance simulating extracellular matrix composition at the early stage of bone regeneration measured by Mo/PLCL electrodes. (A) Impedance of alginate hydrogel with varying concentrations of collagen. **(B)** Impedance of alginate hydrogel with varying concentrations of HA.

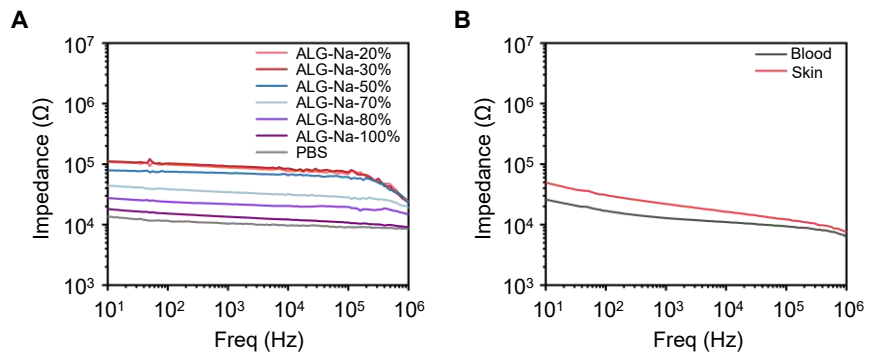


Figure S28. Simulation of the effects of moisture, verification and fibrosis on the impedance of bone tissue. (A) The impedance of sodium alginate with different moisture contents. **(B)** the impedance of rat blood and skin tissue.

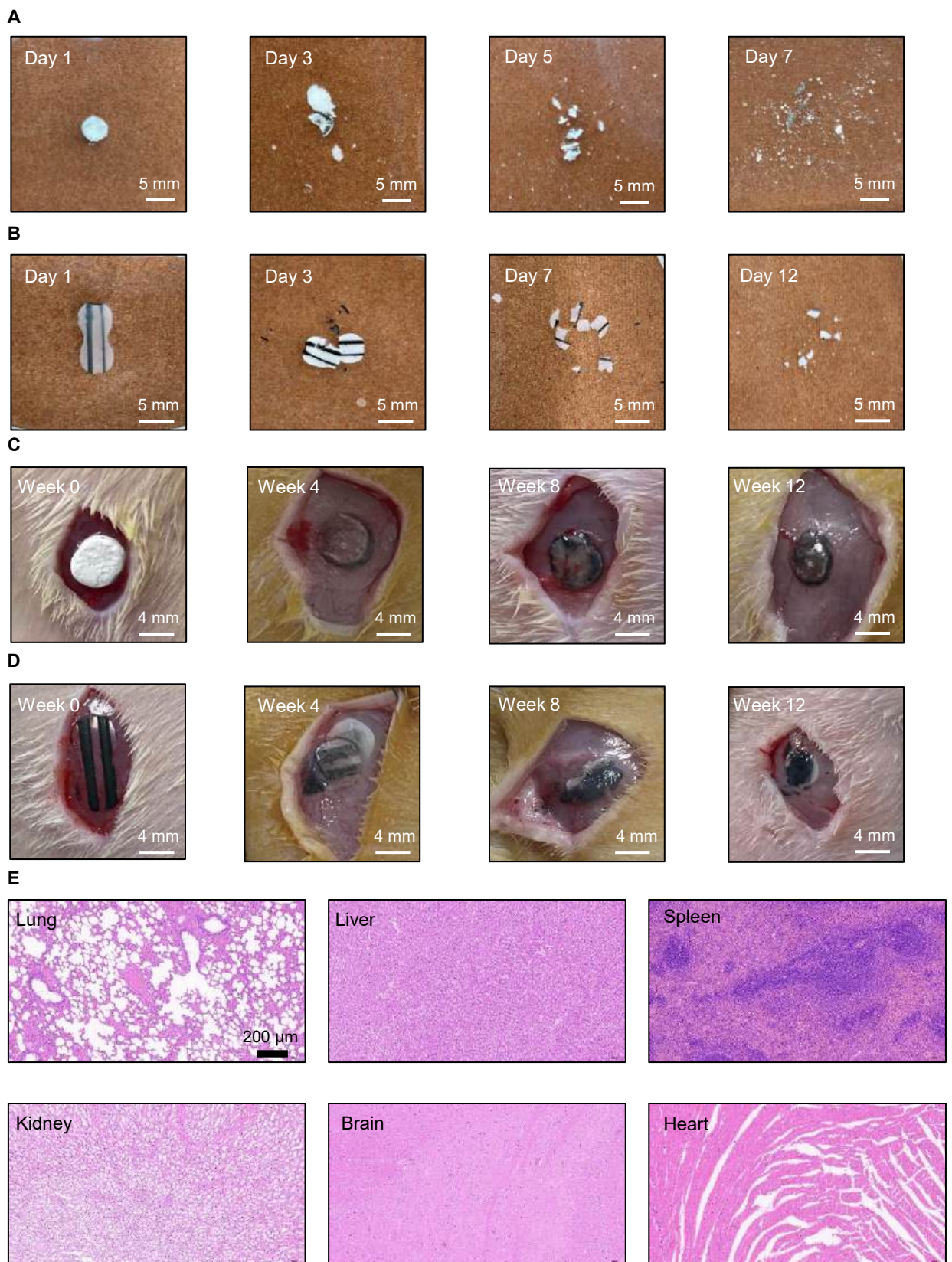


Figure S29. In vivo biodegradation and biocompatibility of the multifunctional cranial device. (A) Images of the self-electrified scaffold at various stages of accelerated degradation during immersion in PBS (pH 7.4) at 60 °C. **(B)** Images of the bioimpedance electrodes at various stages of accelerated degradation during immersion in PBS (pH 7.4) at 60 °C. **(C)** Images of the self-electrified scaffold implanted subdermally in SD rats at various stages of degradation over a 12-week period. **(D)** Images of the bioimpedance electrodes implanted subdermally in SD rats at various stages of degradation over a 12-week period. **(E)** H&E staining of major organs (lung, liver, spleen, kidney, brain and heart) at 12 weeks postimplantation. In **(C–E)** $n = 4$ Independent experiments.